

Hyperbolic Geometry and Hyperbolic GNNs

CPSC483: Deep Learning on Graph-Structured Data

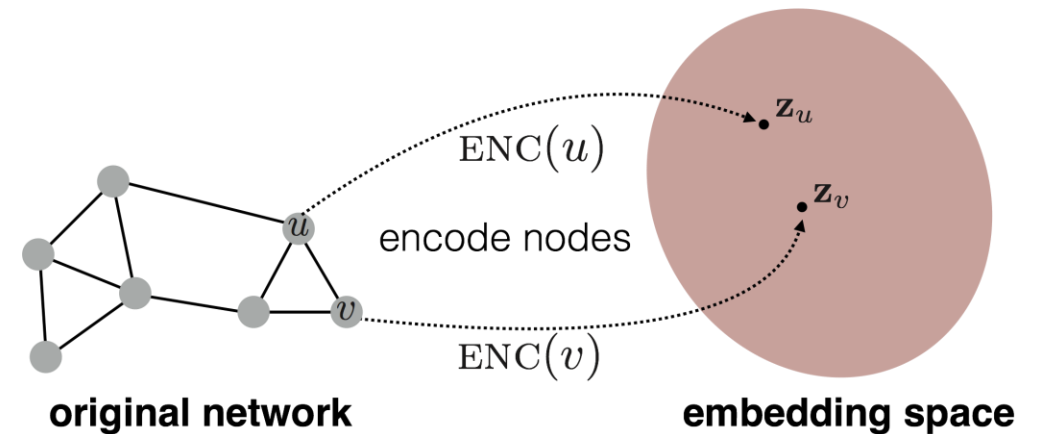
Rex Ying

Readings

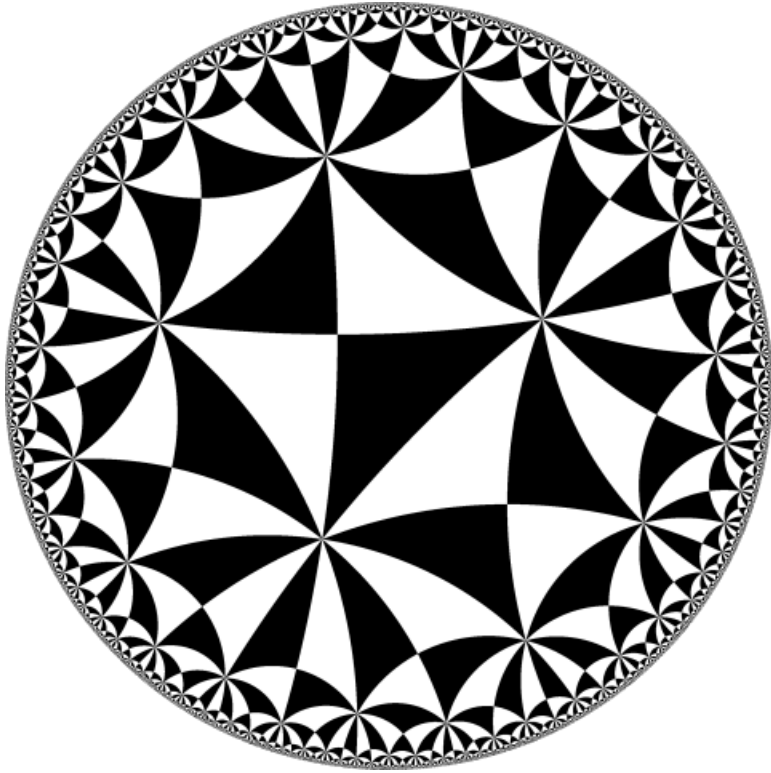
- Readings are updated on the website (syllabus page)
- **Lecture 17 readings:**
 - [HGCN: Hyperbolic Graph Convolutional Neural Networks](#)
 - [Hyperbolic GNN survey](#)

Recap: Graph Representation Learning

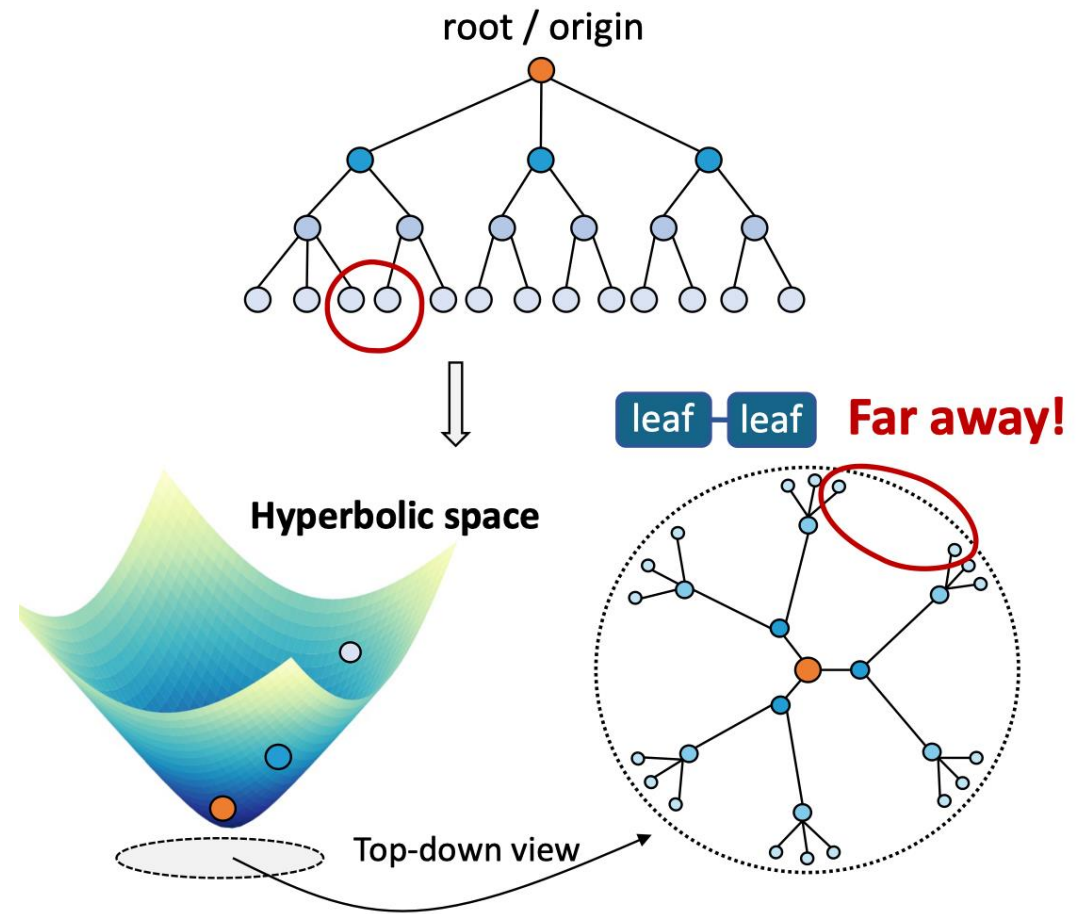
- **Step 1:** Obtain **node and edge features**, possibly augment them with structural properties of the input graphs
- **Step 2:** Use a parameterized **encoder** to map nodes to an embedding space
- **Step 3:** Make **predictions** on nodes/edges/graphs **based on embeddings**
- **Step 4:** Compute **loss** and **optimize** the parameters



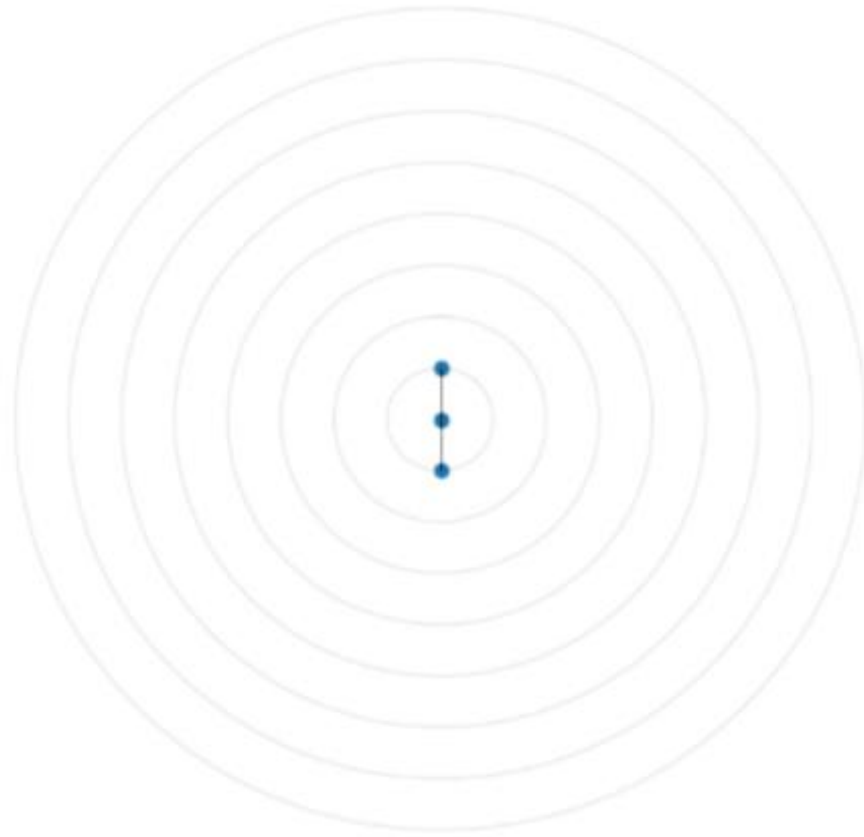
Motivation: Embedding Hierarchical Structure



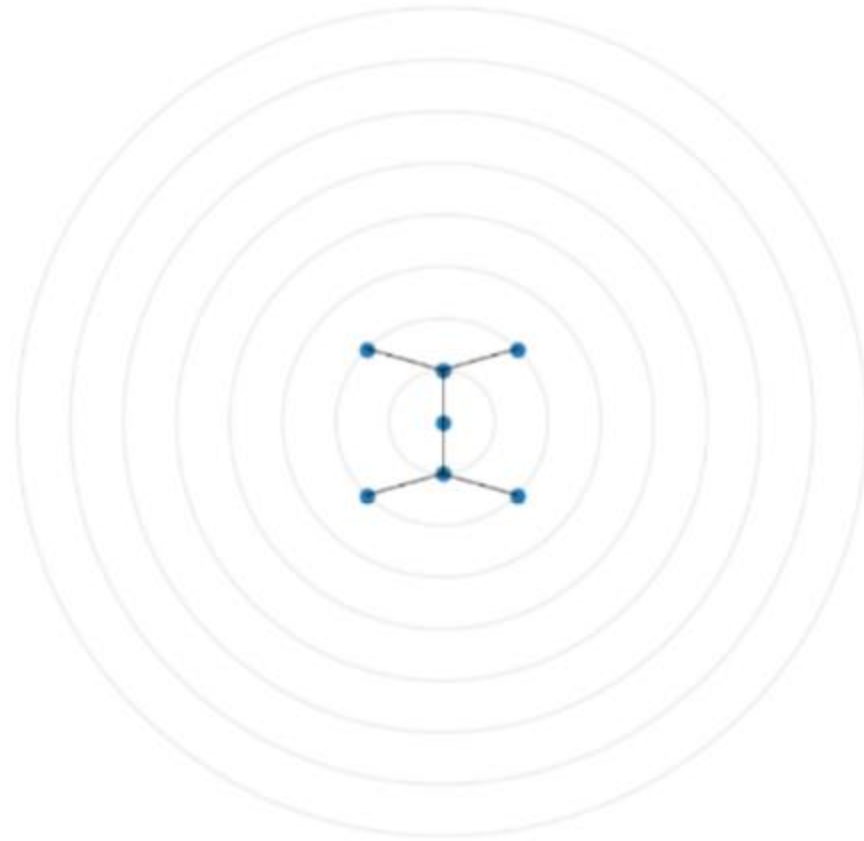
The volume of a ball in the hyperbolic space grows **exponentially** with its radius



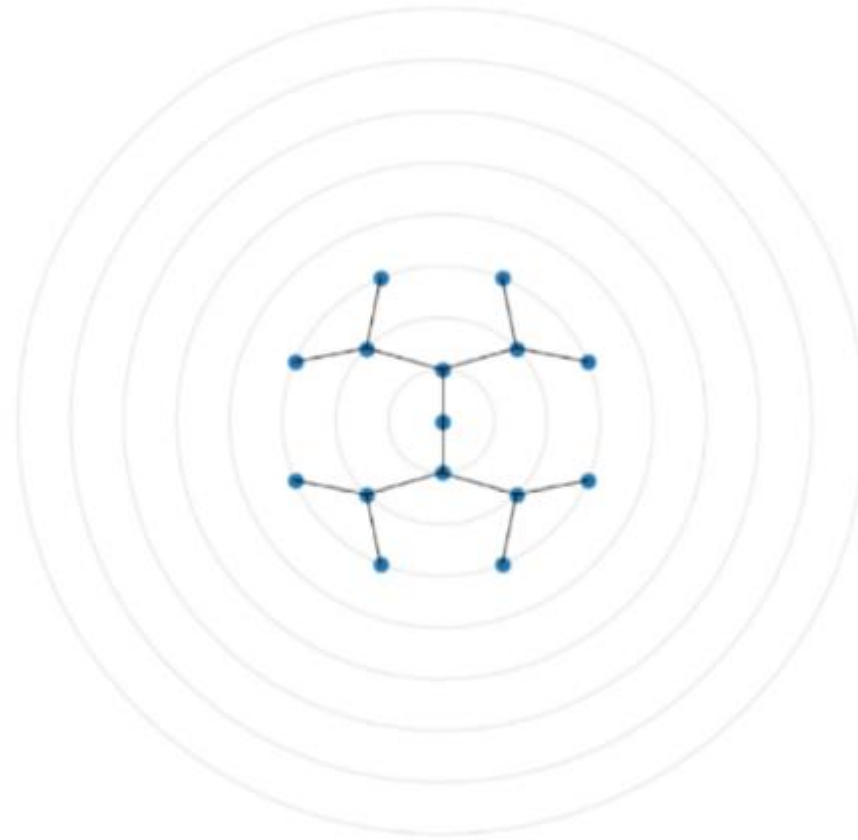
Example: Embedding Tree-structured Data



Example: Embedding Tree-structured Data

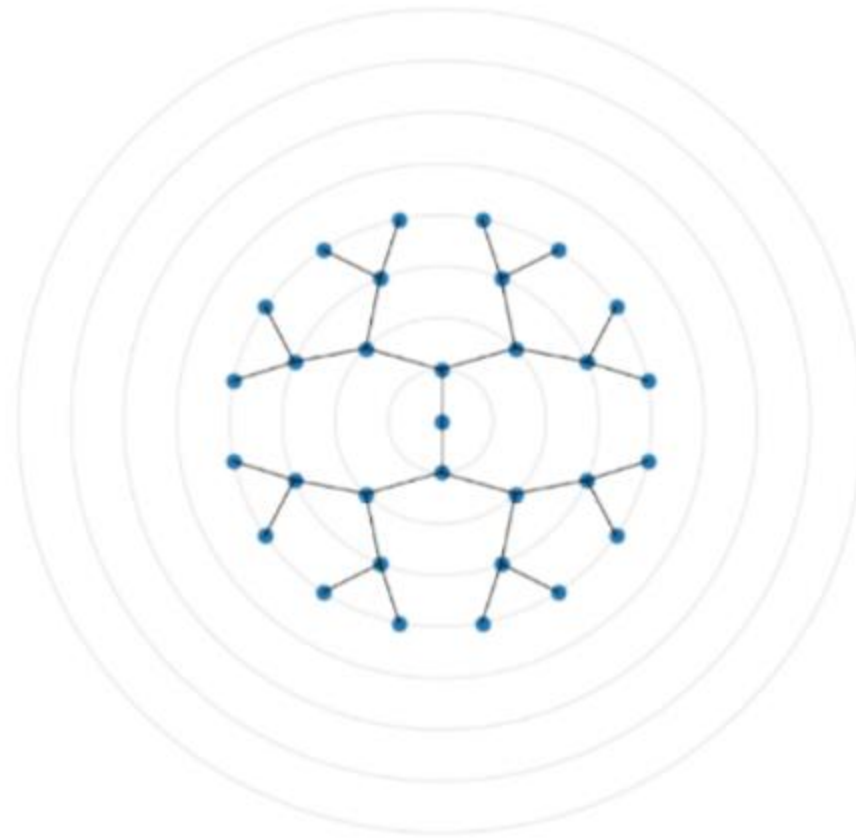


Example: Embedding Tree-structured Data

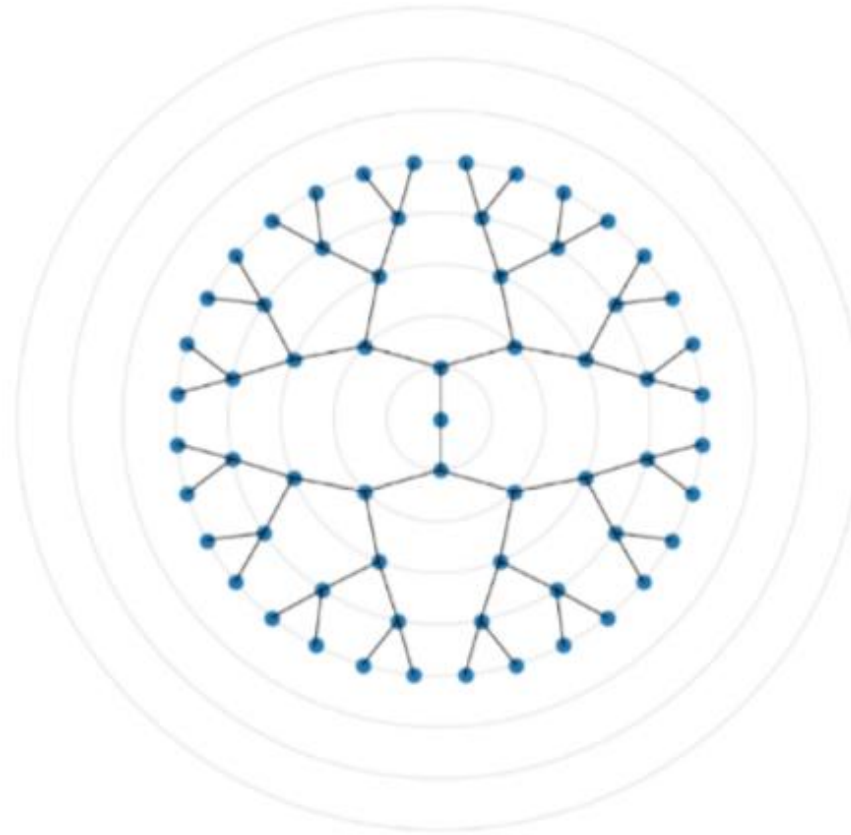


So far, so good
Nodes are close **i.f.f. they
are connected by an edge**

Example: Embedding Tree-structured Data



Example: Embedding Tree-structured Data

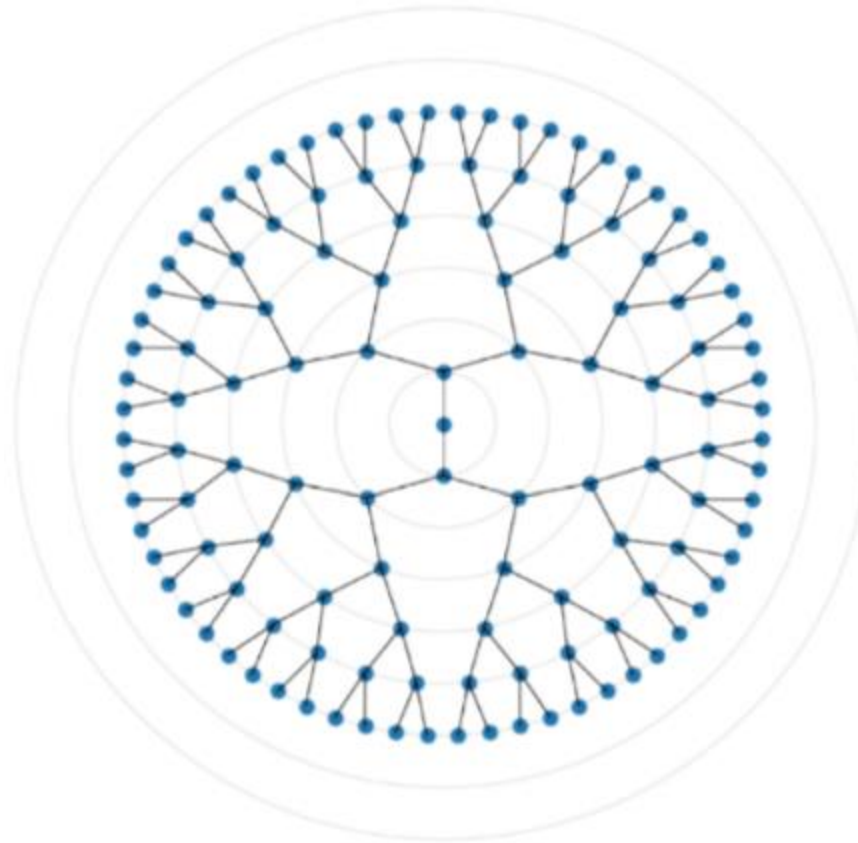


But the outermost nodes are becoming increasingly close to one another.

....

Even though they are not connected by an edge in the graph.

Example: Embedding Tree-structured Data

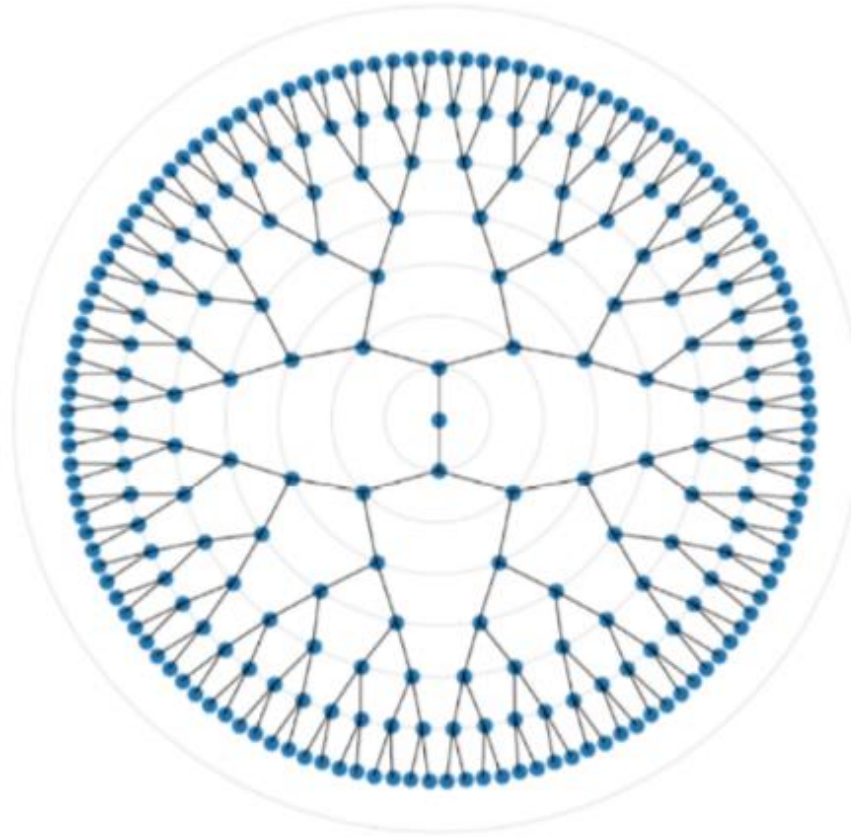


But the outermost nodes are becoming increasingly close to one another.

....

Even though they are not connected by an edge in the graph.

Example: Embedding Tree-structured Data



Things only get worse!
We have lost our
property:

“close i.f.f share edge”

Issues with Euclidean Embeddings: Distortion

- Euclidean space leads to **significant distortion** regardless of the embedding dimensions

Theorem

(Informal; Lee et al., (2007)) There is a lower bound in the minimal distortion of embedding hierarchical structures (e.g. token relationships) into Euclidean space ($\mathbb{R}^n$).

“There is a **performance bottleneck** on how well Euclidean foundation models can represent complex token relationships”

Issues with Euclidean Embeddings: Dimension Dilemma

- Euclidean space face the dilemma of **dimension-distortion tradeoffs**
 - High dimensionality is often required to embed complex token relations in Euclidean space with (relatively) low distortion

Theorem

*(Informal; Matoušek (2002)) The dimension required when embedding unweighted graphs (in the form of token relationships/self-attention) grows **near-quadratically** w.r.t to distortion.*

“Euclidean foundation models have *limited scalability*”

Content

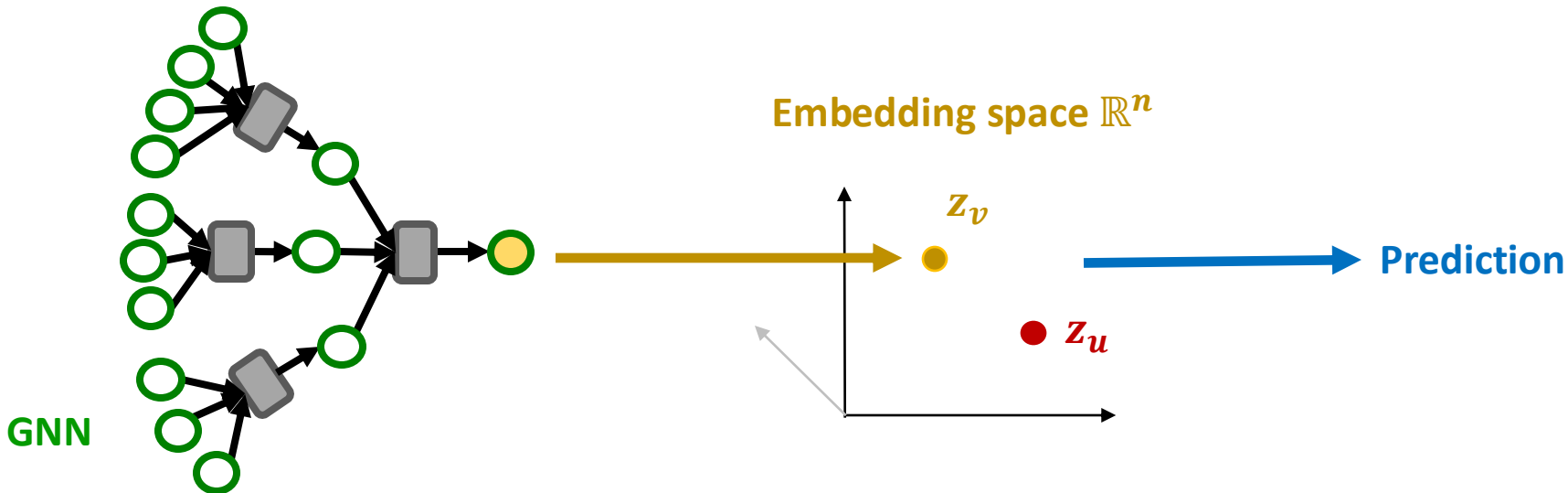
- **Non-Euclidean Space**
- **Hyperbolic Embeddings**
- **Hyperbolic GNNs**

Content

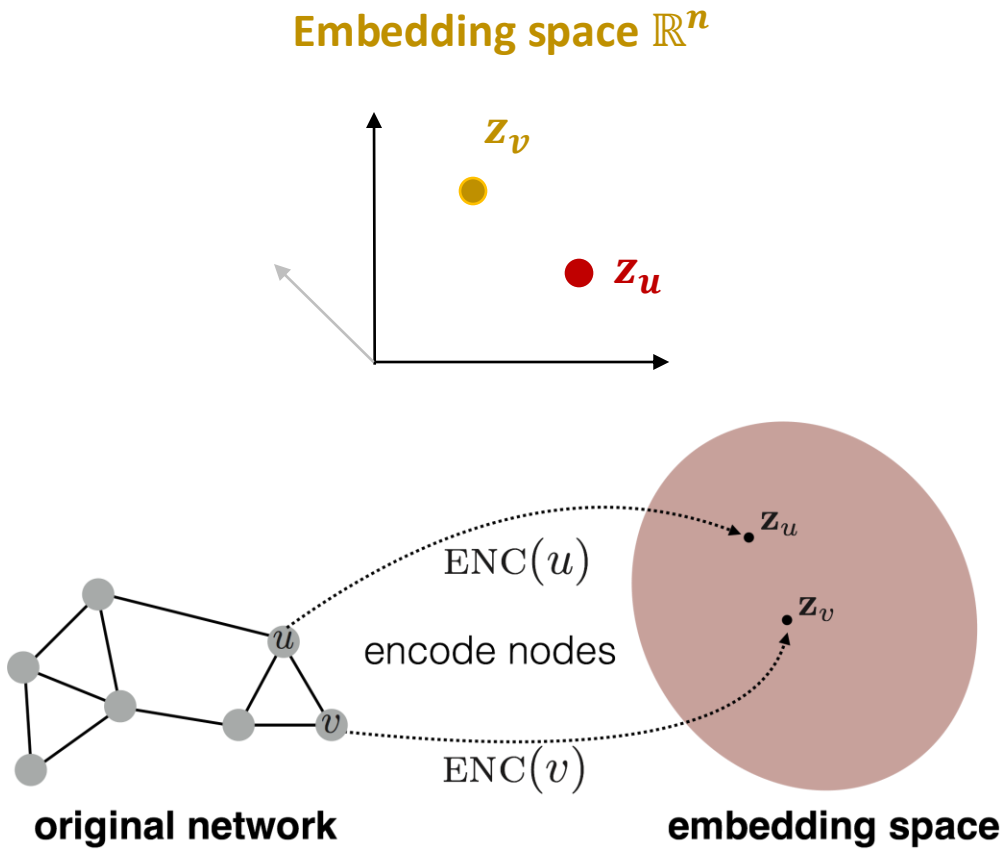
- **Non-Euclidean Space**
- Hyperbolic Embeddings
- Hyperbolic GNNs

Architecture vs. Embedding Geometry

- **Architecture** and **embedding geometry** are both crucial to the expressive power of a neural network
- Embedding geometry is closely related to the objective function
- Better embedding geometry can benefit a variety of architectures



Question



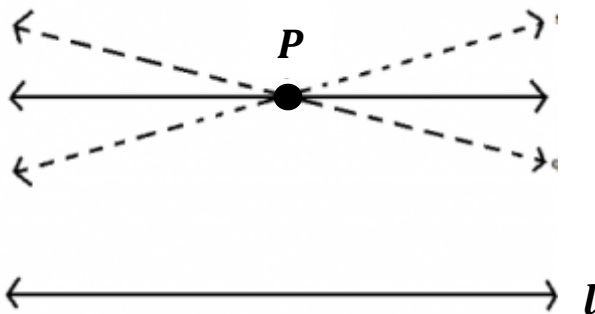
Euclidean Space can not always capture complex graph structures

What embedding space geometry is optimal for data?

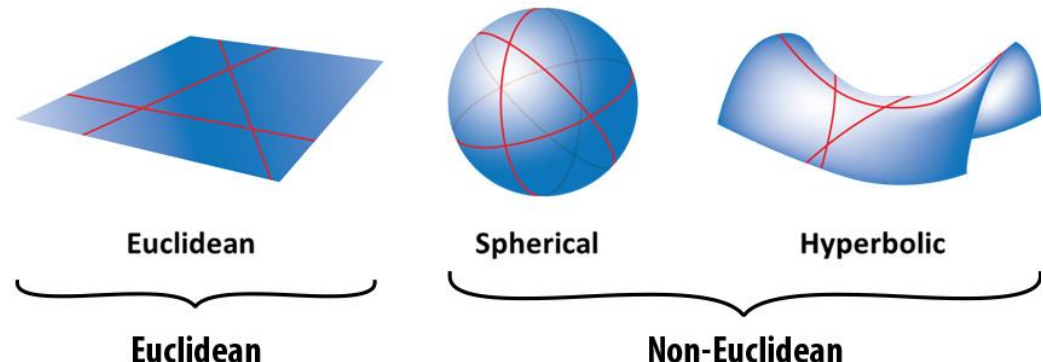
Consider non-euclidean space!

Non-Euclidean Space

- **Euclidean Space** (satisfy the **fifth parallel postulate of Euclidean Geometry**)
 - Given a line and a point not on it, **exactly one line parallel** to the given line can be drawn through the given point.
- **Non-Euclidean Space:**
 - **Hyperbolic:** negative curvature, infinitely many parallel lines (curve away from each other)
 - **Spherical:** positive curvature, no parallel lines

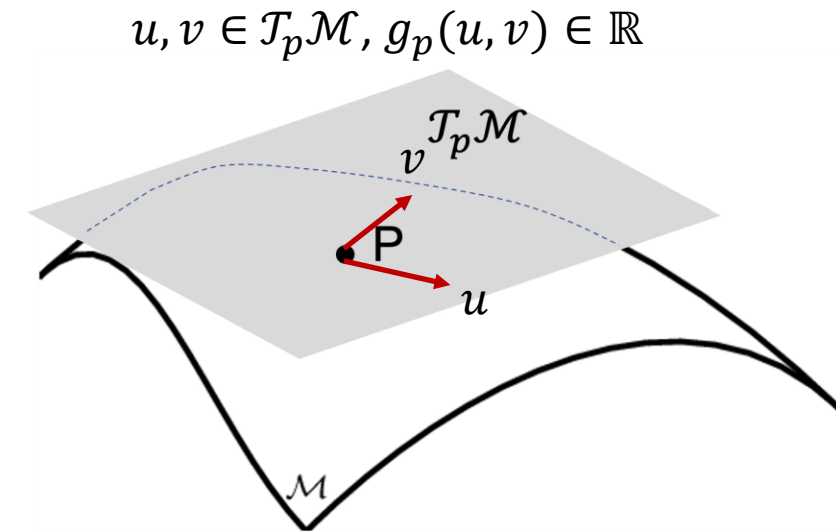


In Euclidean Space,
only one line through
 P that is parallel to l



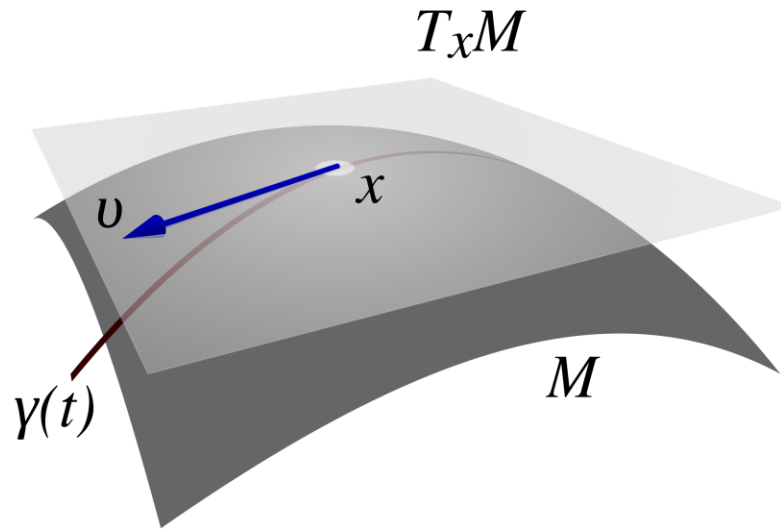
Riemannian Manifold

- **Manifold**: high-dimensional surface
- **Riemannian Manifold $\mathcal{M}$**
 - Equipped with
 - *Tangent space $\mathcal{T}_p\mathcal{M}$* : an $\mathbb{R}^d$ that approximates the manifold at any point $p \in \mathcal{M}$
 - *Inner product g_p* : $\mathcal{T}_p\mathcal{M} \times \mathcal{T}_p\mathcal{M} \rightarrow \mathbb{R}$
 - Both functions vary smoothly (differentiable) on the manifold



Tangent Space

- **Curve:** smooth path along manifold $\gamma: [0,1] \rightarrow \mathcal{M}$
- **Speed:** direction of change along the curve $\dot{\gamma}: [0,1] \rightarrow \mathcal{T}_x\mathcal{M}$
- **Tangent space $\mathcal{T}_x\mathcal{M}$:** space of **speed vectors v** of all curves γ that go through point x on the manifold $\mathcal{M}$

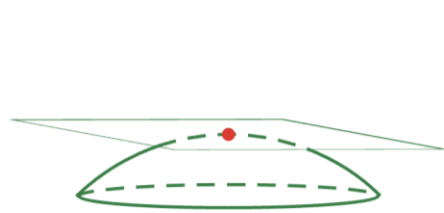


Curvature

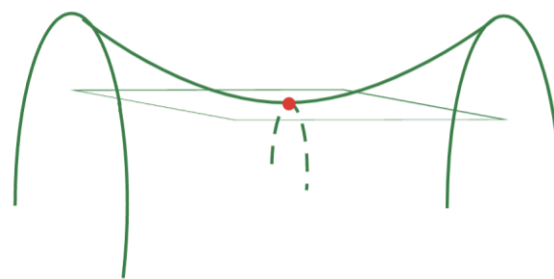
- The **curvature** (sectional curvature) at a point measures how drastically a surface **bends away** from its tangent plane at this point

High-level Intuition:

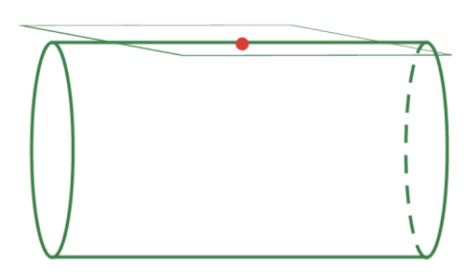
- If the surface locally lives **entirely on one side** of the tangent space $\mathcal{T}_p\mathcal{M} \Rightarrow$ **Positive** curvature at point p
- If the tangent space $\mathcal{T}_p\mathcal{M}$ **cuts through** the surface $\Rightarrow$ **Negative** curvature at point p
- If the surface has a line along which the **surface agrees with the tangent space** $\mathcal{T}_p\mathcal{M} \Rightarrow$ **Zero** curvature at point p



positive curvature



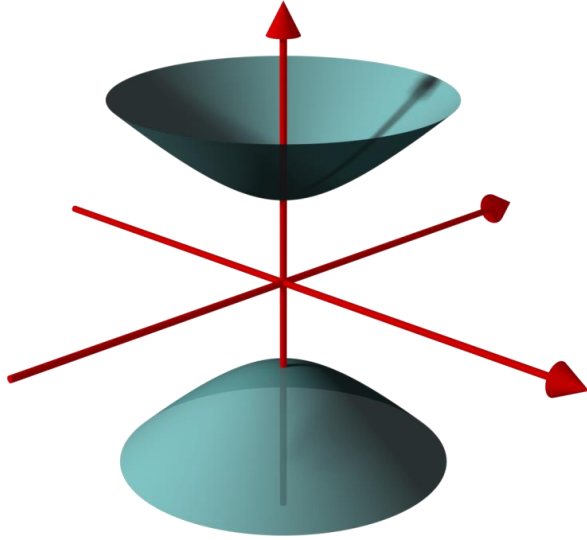
negative curvature



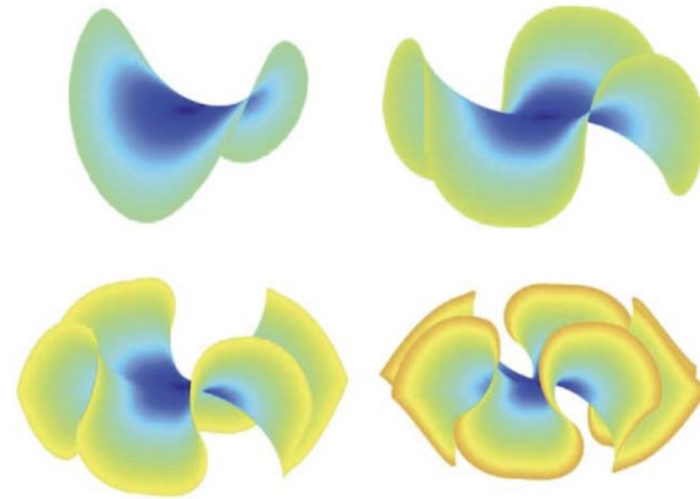
zero curvature

Hyperbolic Space

- **Hyperbolic space** is a Riemannian manifold with **constant negative curvature** $-1/K$, where $(K > 0)$
 - Becomes Euclidean when $K \rightarrow \infty$
- In **Euclidean space**, we can also find manifolds with constant negative curvature:



two sheet hyperboloid (source: Wikipedia)



[Periodic Amsler Surfaces](#)

Hyperbolic Space and Minkowski Space

Hilbert's Theorem (1901): There exists no hyperbolic surface with constant negative curvature that can be embedded into $\mathbb{R}^3$.

- However, we can embed hyperbolic geometry into **Minkowski Space**
- The **Minkowski metric** in the Minkowski space is different from the Euclidean metric.
 - **Euclidean Metric**: $g_E(\mathbf{u}, \mathbf{v}) = u_0 v_0 + u_1 v_1 + \cdots + u_d v_d$
 - **Minkowski Metric**: $g_M(\mathbf{u}, \mathbf{v}) = \pm(u_0 v_0 - u_1 v_1 - \cdots - u_d v_d)$
 - Without loss of generality we can take the + sign
 - Note: dimension 1 is treated differently in Minkowski Space.

Inner Product

- **Hyperboloid model** as a Riemannian manifold:

- With Constant **Minkowski metric**:

$$\langle \cdot, \cdot \rangle_{\mathcal{L}} : \mathbb{R}^{d+1} \times \mathbb{R}^{d+1} \rightarrow \mathbb{R}$$

$$\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}} = \boxed{-x_0 y_0} + \boxed{x_1 y_1 + \dots + x_d y_d}$$

Time-like Space-like



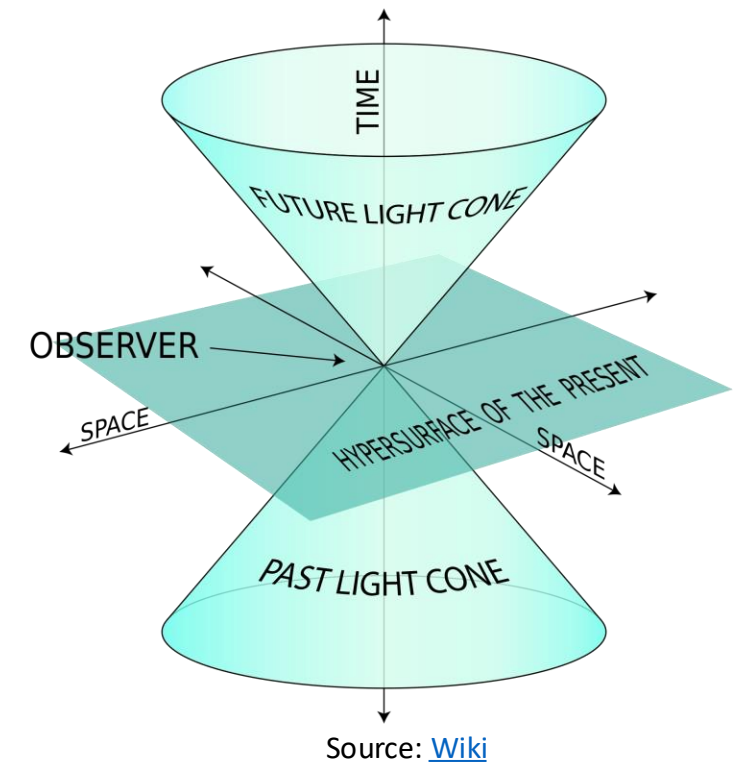
- **Hyperboloid model** $\mathbb{H}^{d,K} = \{\mathbf{x} \in \mathbb{R}^{d+1} : \langle \mathbf{x}, \mathbf{x} \rangle_{\mathcal{L}} = -K\}$, $-\frac{1}{K}$ is the curvature
- **Note:** the points in hyperboloid model $\mathbb{H}^{d,K}$ are represented in $(d + 1)$ -dimensional Minkowski space.
- The metric of hyperboloid model is different from the Euclidean metric!

Minkowski Space and Special Relativity

- Minkowski space-time (4D) : $g_M(p_1, p_2) = t_1 t_2 - x_1 x_2 - y_1 y_2 - z_1 z_2$
- A vector is **time-like** if $\|(t_x, x_1 \dots, x_d)\| > 0$
- A vector is **space-like** if $\|(t_x, x_1 \dots, x_d)\| < 0$
- A vector is **light-like** if $\|(t_x, x_1 \dots, x_d)\| = 0$
 - Defines a light cone
- Lorentz transformation can be viewed as **rotation** in a hyperbolic space
 - Lorentz boost (transformation along an axis)

$$\begin{array}{ccc} \begin{array}{l} t' = \gamma \left(t - \frac{vx}{c^2} \right) \\ x' = \gamma (x - vt) \\ y' = y \\ z' = z \end{array} & \xrightarrow{\text{Hyperbolic rotation along x-t plane}} & \begin{array}{l} ct' = ct \cosh \zeta - x \sinh \zeta \\ x' = x \cosh \zeta - ct \sinh \zeta \\ y' = y \\ z' = z \end{array} \end{array}$$

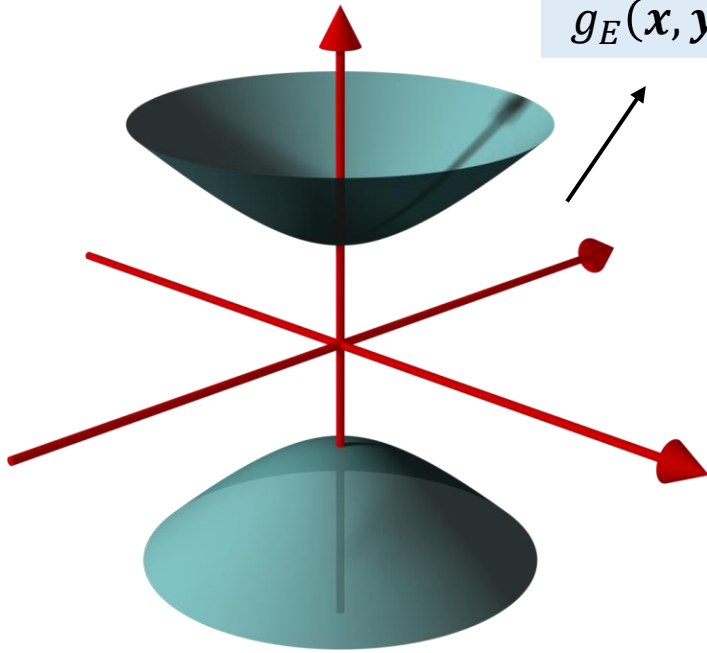
γ : Lorentz factor



Hyperboloid in Different Spaces

Euclidean Metric

$$g_E(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2 + x_3y_3$$



two sheet hyperboloid in 3D Euclidean space

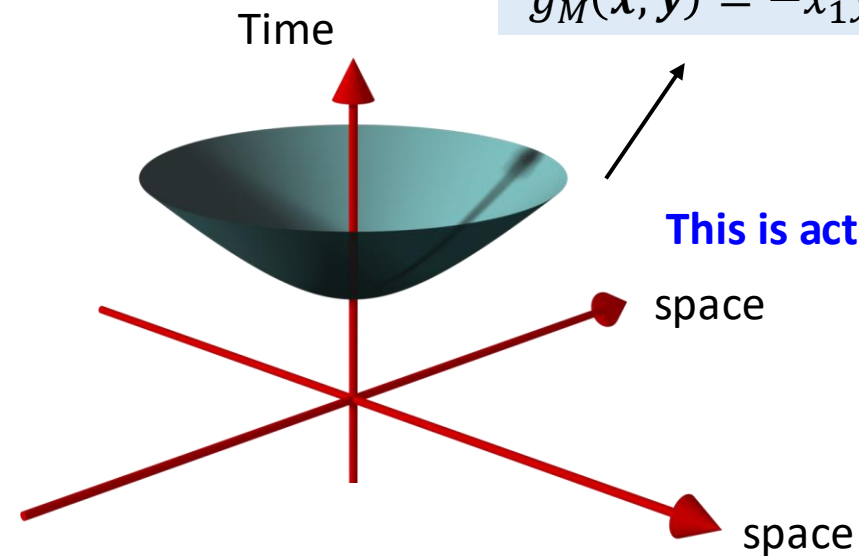
Geodesic distance in Euclidean hyperboloid:

$$d_E(\mathbf{x}, \mathbf{y}) = \sqrt{2(1 - g_E(\mathbf{x}, \mathbf{y}))}$$

(with normalized $\mathbf{x}$ and $\mathbf{y}$)

Minkowski Metric

$$g_M(\mathbf{x}, \mathbf{y}) = -x_1y_1 + x_2y_2 + x_3y_3$$



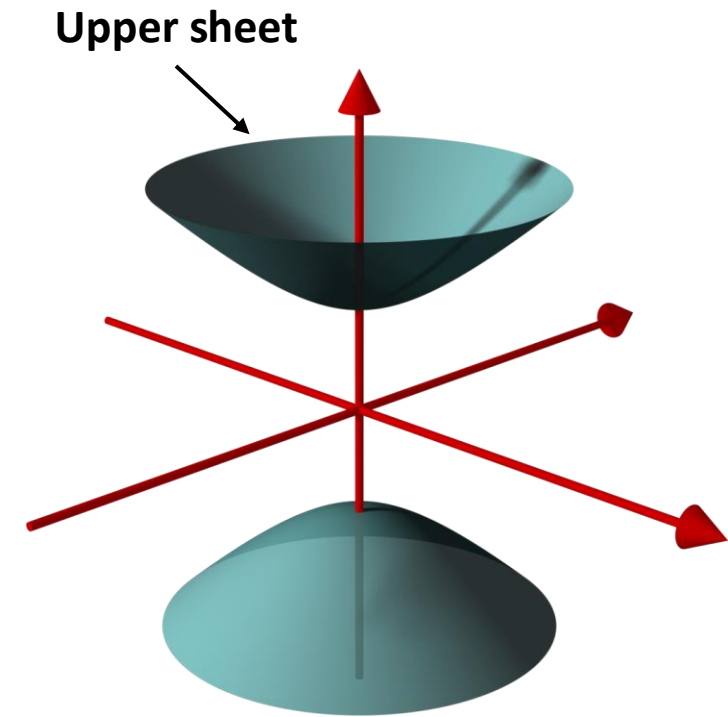
2D Hyperboloid model in 3D Minkowski space

Geodesic distance in Minkowski hyperboloid:

$$D_M^K(\mathbf{x}, \mathbf{y}) = \sqrt{K} \operatorname{arcosh}\left(-\frac{g_M(\mathbf{x}, \mathbf{y})}{K}\right)$$

Hyperboloid Model

- **Hyperboloid Model (Lorentz Model)**
 - Upper sheet of 2-sheet hyperboloid
 - d -dimensional Hyperboloid can be represented in $(d + 1)$ -dimensional Minkowski space
 - Subset of Euclidean space
 - Numerically more stable, simpler formula

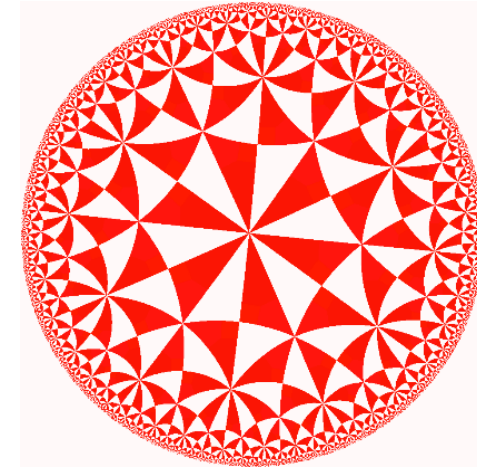


two sheet hyperboloid

Poincaré Model

- **Poincaré Model**

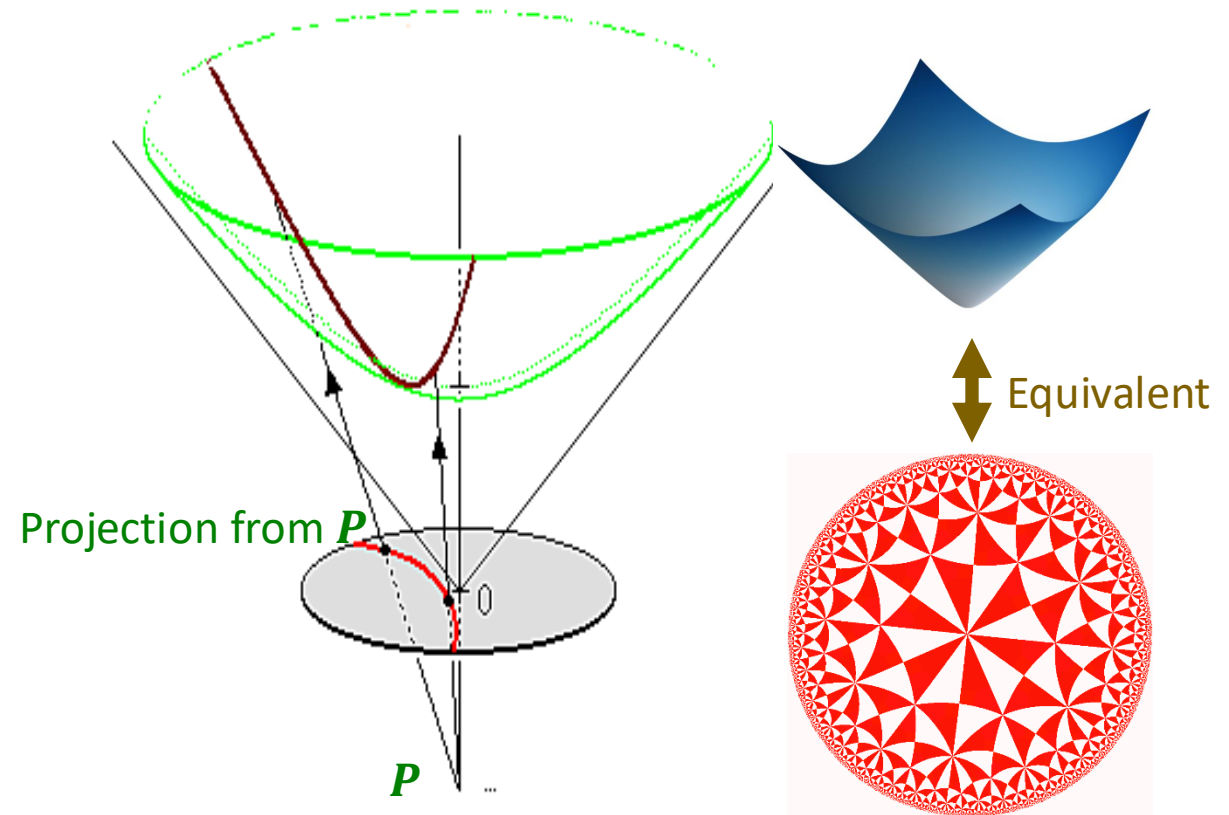
- Radius proportional to $\sqrt{K}$ ($-\frac{1}{K}$ is the curvature)
- Open ball (exclude boundary)
- Each triangle in the figure has the **same** area
- **Exponentially many triangles** with the same area towards the boundary of **Poincaré Ball**



Poincaré: intuitive visualization

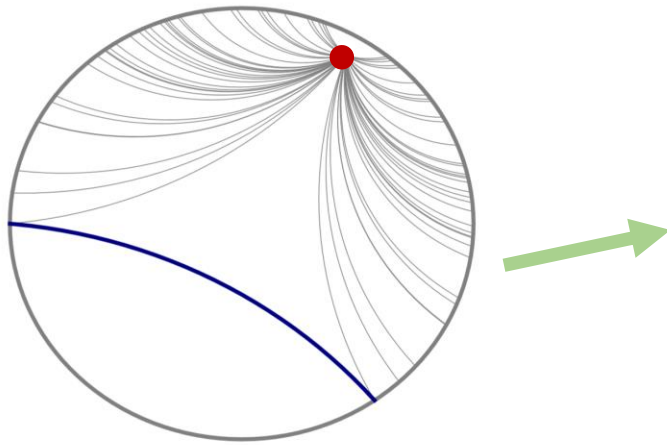
Equivalence

- d -dimensional Poincaré model and $(d + 1)$ -dimensional hyperboloid model are **equivalent**!
- 2d Poincaré model can be derived using a **projection** of 3d hyperboloid model through a specific point onto the unit circle of the $z = 0$ plane.



Geodesic

- **Geodesic:** shortest path in manifold
 - Analogous to straight lines in $\mathbb{R}^n$
 - Curved in hyperbolic space
- Geodesics visualization in Poincaré model: curved!



Set of geodesic lines from the **red** point to boundary of the Poincare ball that are parallel to the **blue line**

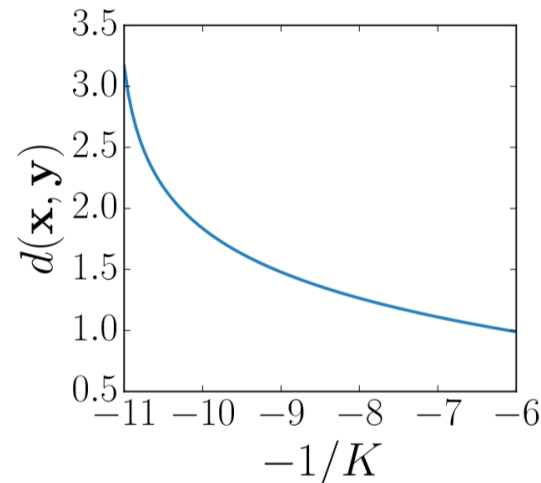
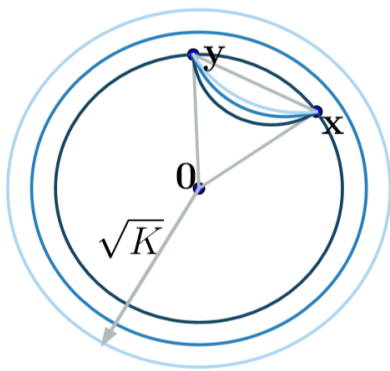
Geodesic Distance

- **Geodesic distance** between $\mathbf{x}$ and $\mathbf{y}$ for $\mathbb{H}^{d,K}$:

$$D_{\mathcal{L}}^K(\mathbf{x}, \mathbf{y}) = \sqrt{K} \operatorname{arcosh}\left(-\frac{\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}}}{K}\right)$$

- The **more negative** the curvature:
 - the more geodesics bends **inward**
 - geodesic **distance increases**

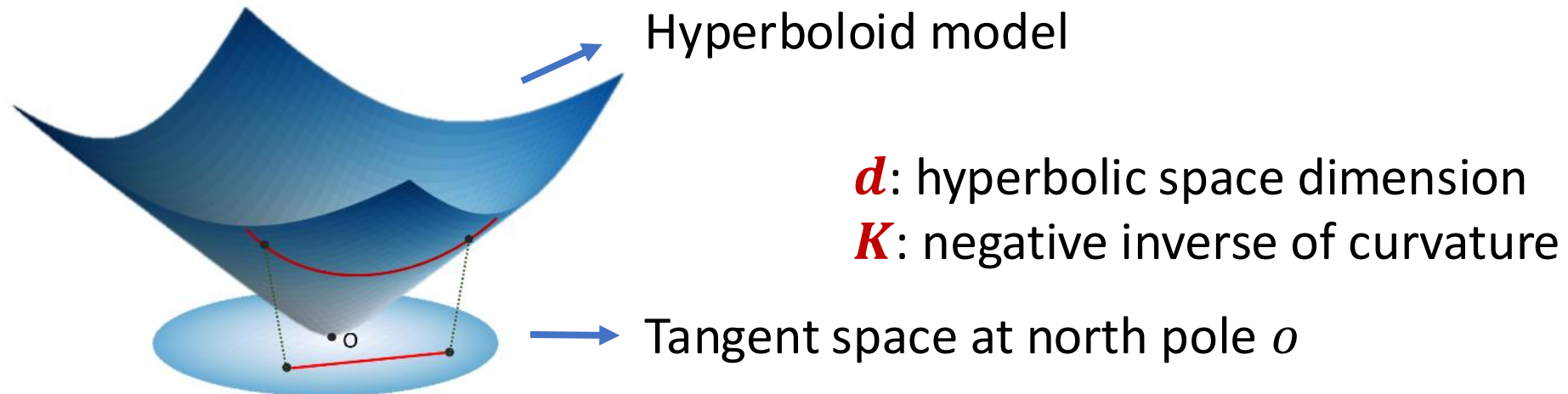
$$\operatorname{arcosh}(x) = \ln(x + \sqrt{x^2 + 1})$$



Dark blue: high curvature boundary and geodesics
Light blue: low curvature boundary and geodesics

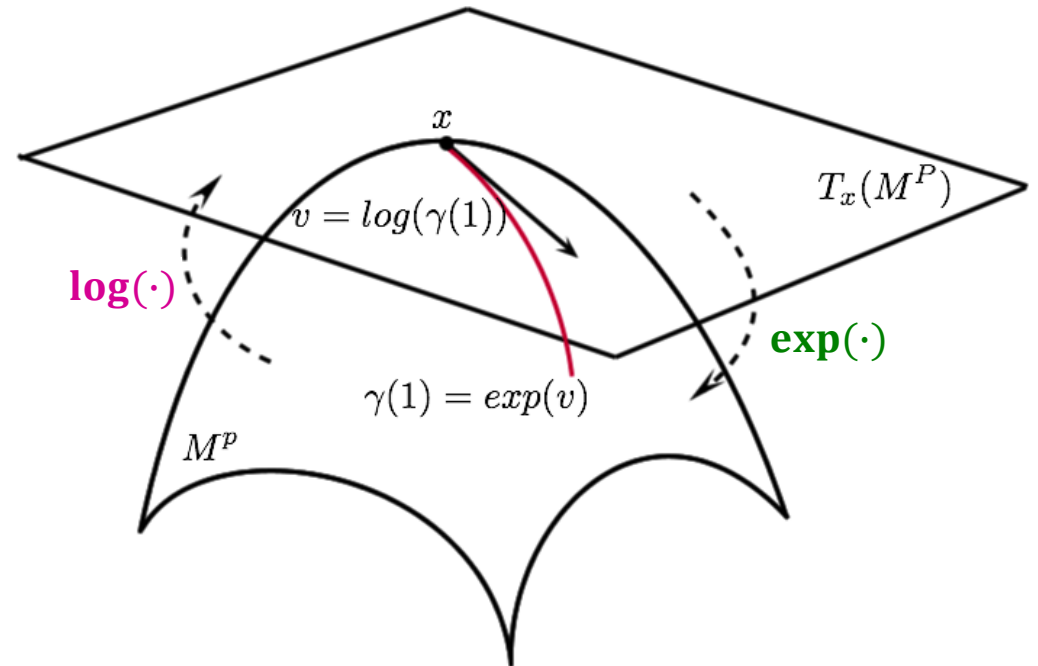
Tangent Space

- Tangent space expression under **hyperboloid model** $\mathbb{H}^{d,K}$ at point $\mathbf{x}$:
 - $\mathcal{T}_{\mathbf{x}}\mathbb{H}^{d,K} = \{\mathbf{v} \in \mathbb{R}^{d+1} : \langle \mathbf{v}, \mathbf{x} \rangle_{\mathcal{L}} = 0\}$
- A vector space (linear structure) with **the same dimension as the hyperboloid model**
- The best **linear approximation** to the manifold $\mathbb{H}^{d,K}$ at point $\mathbf{x}$



Mapping to and from Tangent Space

- **Exponential map:** $\mathcal{T}_x \mathbb{H}^{d,K} \rightarrow \mathbb{H}^{d,K}$
 - from tangent space (Euclidean) to manifold
- **Logarithmic map:** $\mathbb{H}^{d,K} \rightarrow \mathcal{T}_x \mathbb{H}^{d,K}$
 - from manifold to tangent space
 - inverse operation of exponential map

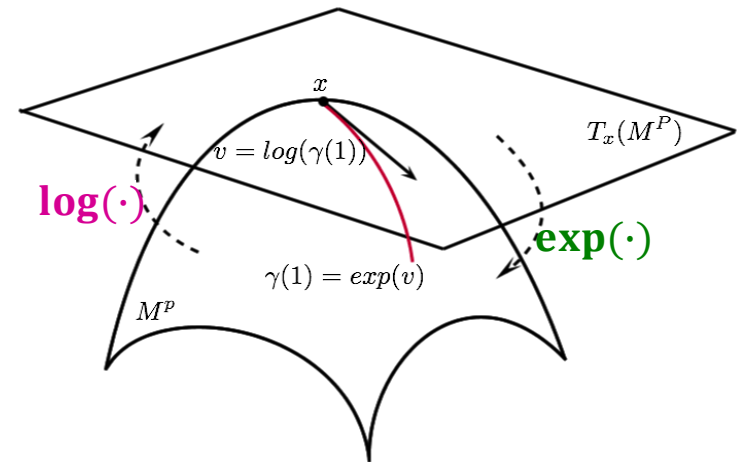


Exponential Map:

- For **hyperboloid model** $\mathbb{H}^{d,K} = \{x \in \mathbb{R}^{d+1} : \langle x, x \rangle_{\mathcal{L}} = -K\}$ at point x
- **Exponential Map:**

$$\exp_x^K(v) = \cosh\left(\frac{\|v\|_{\mathcal{L}}}{\sqrt{K}}\right) x + \sqrt{K} \sinh\left(\frac{\|v\|_{\mathcal{L}}}{\sqrt{K}}\right) \frac{v}{\|v\|_{\mathcal{L}}}$$

- $v \in \mathcal{T}_x \mathbb{H}^{d,K}$
- $\cosh(x) = \frac{e^x + e^{-x}}{2}$, $\sinh(x) = \frac{e^x - e^{-x}}{2}$
- $\|v\|_{\mathcal{L}} = \langle v, v \rangle_{\mathcal{L}}$

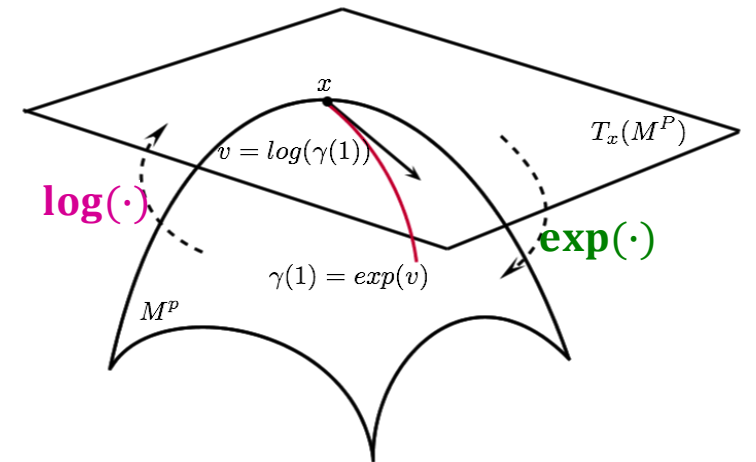


Logarithmic Map

- For **hyperboloid model** $\mathbb{H}^{d,K} = \{\mathbf{x} \in \mathbb{R}^{d+1} : \langle \mathbf{x}, \mathbf{x} \rangle_{\mathcal{L}} = -K\}$ at point $\mathbf{x}$
- **Logarithmic map:**

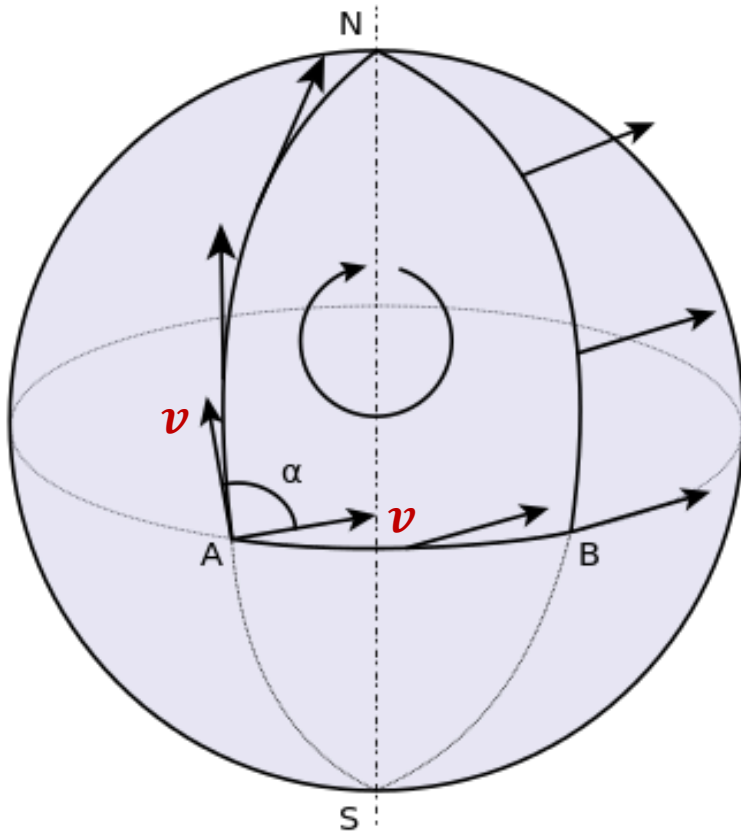
$$\log_x^K \mathbf{y} = D_{\mathcal{L}}^K(\mathbf{x}, \mathbf{y}) \frac{\mathbf{y} + \frac{1}{K} \langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}} \mathbf{x}}{\left\| \mathbf{y} + \frac{1}{K} \langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}} \mathbf{x} \right\|_{\mathcal{L}}}$$

- $\mathbf{y} \in \mathbb{H}^{d,K}$
- $D_{\mathcal{L}}^K(\mathbf{x}, \mathbf{y}) = \sqrt{K} \operatorname{arccosh}\left(-\frac{\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}}}{K}\right)$ is geodesic distance



Parallel Transport (1)

- **Parallel Transport:** transport a vector along a smooth curve on the surface and keep parallel to itself locally.



Transport a tangent vector v along the surface **with non-zero curvature**. When travelling from A to N to B back to A, the direction of the vector v changes!

Parallel Transport (2)

- **Parallel Transport** $P_{x \rightarrow y}(\cdot)$ maps a vector $\mathbf{v} \in \mathcal{T}_x \mathcal{M}$ to $P_{x \rightarrow y}(\mathbf{v}) \in \mathcal{T}_y \mathcal{M}$
- If two points $\mathbf{x}$ and $\mathbf{y}$ on the hyperboloid $\mathbb{H}^{d,K}$ are **connected by a geodesic**, then the parallel transport of tangent vector $\mathbf{v} \in \mathcal{T}_x \mathbb{H}^{d,K}$ to $\mathcal{T}_y \mathbb{H}^{d,K}$:

$$P_{x \rightarrow y}(\mathbf{v}) = \mathbf{v} - \frac{\langle \log_x^K(\mathbf{y}), \mathbf{v} \rangle_{\mathcal{L}}}{D_{\mathcal{L}}^K(\mathbf{x}, \mathbf{y})^2} (\log_x^K \mathbf{y} + \log_y^K \mathbf{x})$$

- $\log_x^K$ is the **Logarithmic map** at point $\mathbf{x}$.
- $D_{\mathcal{L}}^K(\mathbf{x}, \mathbf{y}) = \sqrt{K} \operatorname{arccosh}(-\frac{\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}}}{K})$ is geodesic distance

Content

- Non-Euclidean Space
- **Hyperbolic Embeddings**
- Hyperbolic GNNs

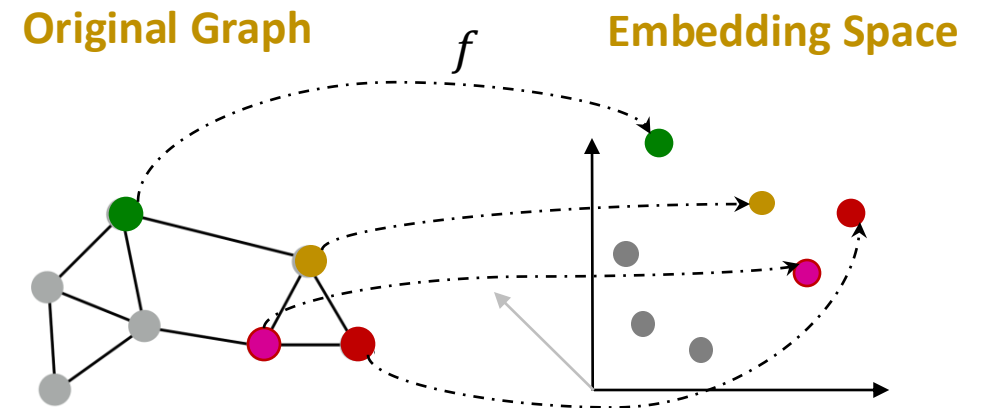
Optimal Embedding

Given a graph $G(V, E)$. Mapping $f: V \rightarrow W$, with distances d_V and d_W

How to measure the **quality of embedding**?

High-level Intuition:

- Consider node $i \in V$, the embeddings of neighbor node in $\mathcal{N}(i)$ should be close to $f(i)$ in the embedding space W
- Distances between embedding vectors $f(i)$ and $f(j)$ in the embedding space W should be close to the distance in original graph G
 - Recall Position-aware GNNs (lecture 10)

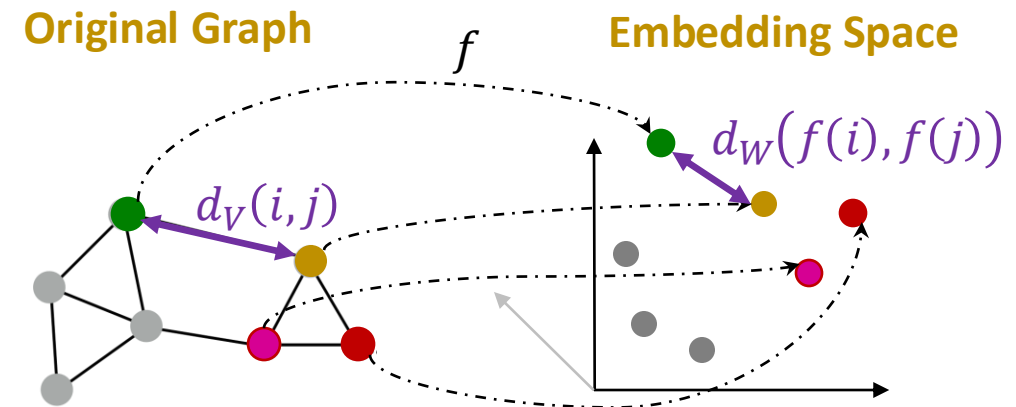


Distance Distortion

- **Distance Distortion:**

$$D(f) = \frac{1}{C_n^2} \left(\sum_{i,j \in V, i \neq j} \frac{|d_W(f(i), f(j)) - d_V(i, j)|}{d_V(i, j)} \right)$$

- $C_n^2 = \frac{n(n-1)}{2}$
- The lower distortion, the better embedding
- The best distortion is $D(f) = 0$, preserving the distances between node pairs exactly

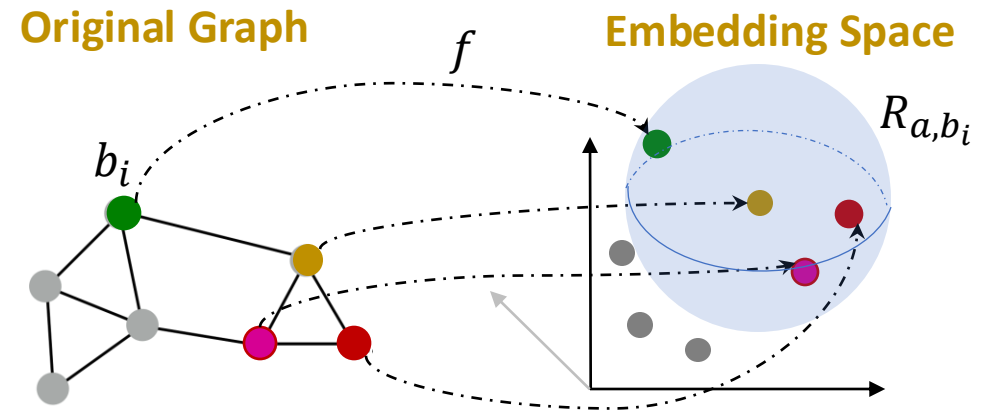


Mean Average Precision

- **Mean Average Precision (mAP)**

$$mAP(f) = \frac{1}{|V|} \sum_{a \in V} \frac{1}{\deg(a)} \sum_{b_i \in \mathcal{N}_a} \frac{|\mathcal{N}_a \cap R_{a,b_i}|}{|R_{a,b_i}|}$$

- V is the node set, $\deg(a)$ denotes the degree of node a
- $\mathcal{N}_a$ denotes the 1-hop neighbor nodes of a
- R_{a,b_i} is the set of nodes whose embeddings fall into the smallest ball centered at the embedding of a , that can retrieve b_i
- Used [here](#) at page 3
- The larger MAP, the better embedding.
- $MAP(f) \leq 1$
- Note: we do not consider node features here

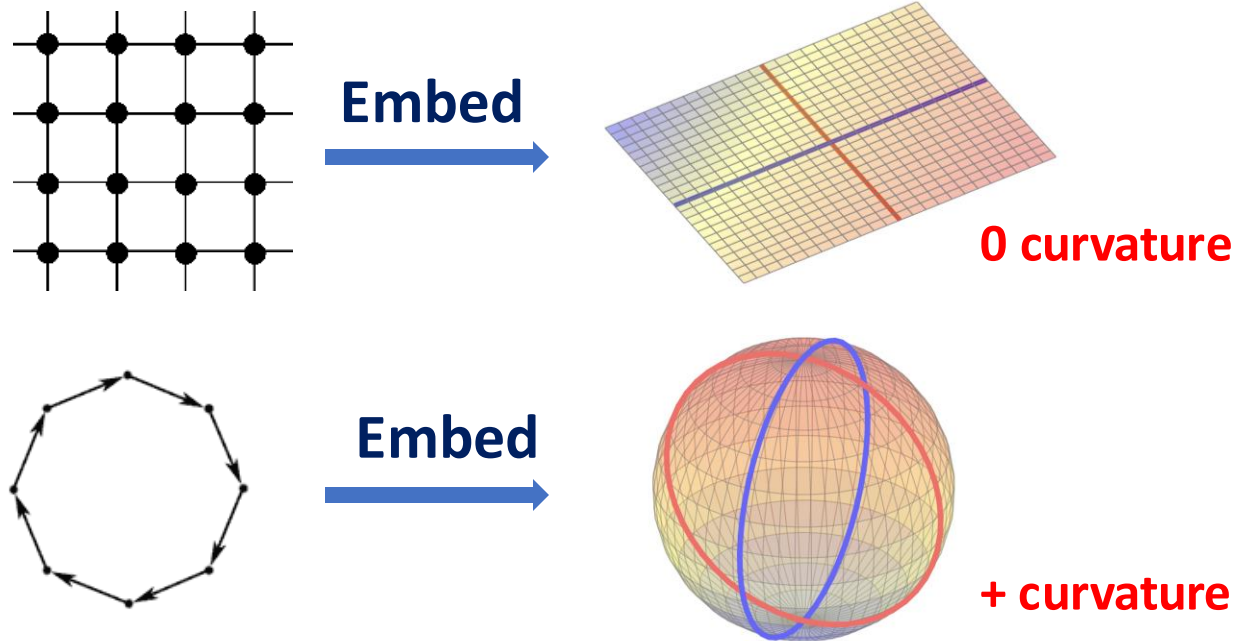


Distortion vs. mAP

- **mAP** is a local measurement which does not depend on an explicit distance
 - It combines the effect of **precision** and **recall** when performing link prediction task
- **Distortion** is a global metric and helps to preserve the explicit value of distances
 - It can be useful in applications where we need to approximate more complex distances than link prediction (which can be viewed as a binary version of distance)
 - Examples: graph / sequence edit distance, shortest path distance, transportation distance (e.g., Google Map) ...

Graph with Grids and Cycles

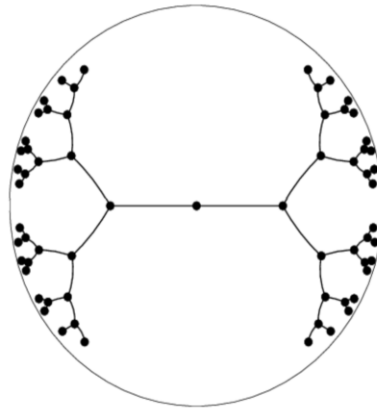
- **Euclidean** space preserves neighbor nodes and distances for **grid-like** graphs
- **Spherical** space preserves neighbor nodes and distances for **cycle-like** graphs
- $\text{mAP}(f) = 1, D(f) = 0$



Graph with Grids and Cycles

- **Euclidean** space preserves neighbor nodes and distances for **grid-like** graphs
- **Spherical** space preserves neighbor nodes and distances for **cycle-like** graphs
- $\text{mAP}(f) = 1, D(f) = 0$

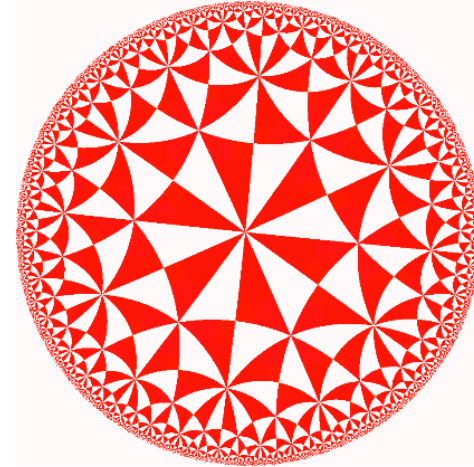
What about Tree-like graphs?



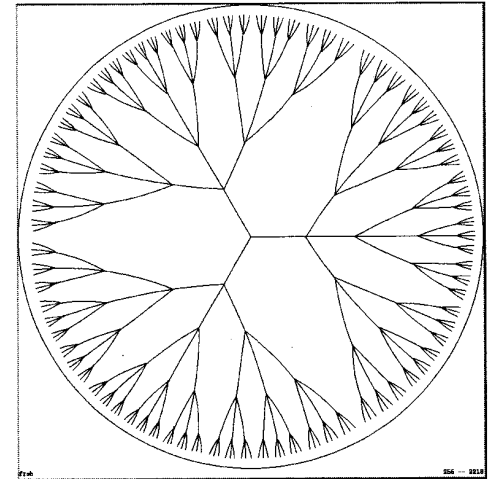
Hyperbolic Geometry!

Exponential volume growth

- The **volume** of d -dimensional Euclidean Ball with radius r :
$$V_d^E(r) \propto r^d$$
- In a tree, the number of nodes **grows exponentially with the tree depth**
- The volume of a **Poincaré model** in the hyperbolic space grows exponentially with its radius!
$$V_2^H(r) \propto e^r$$



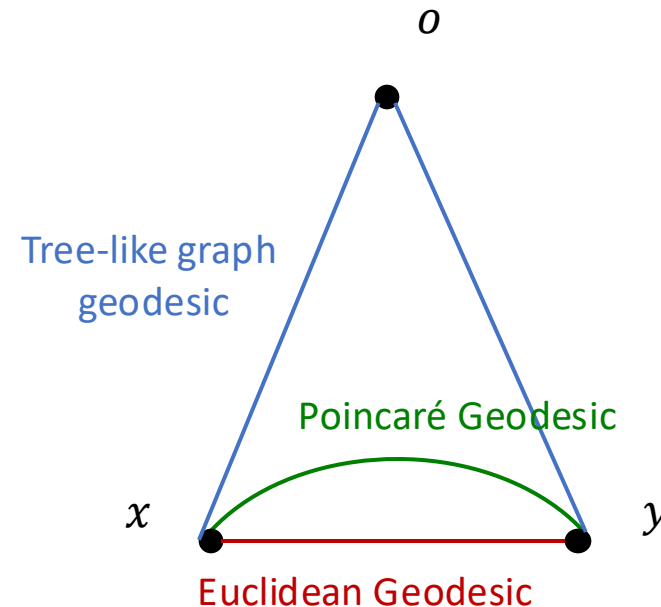
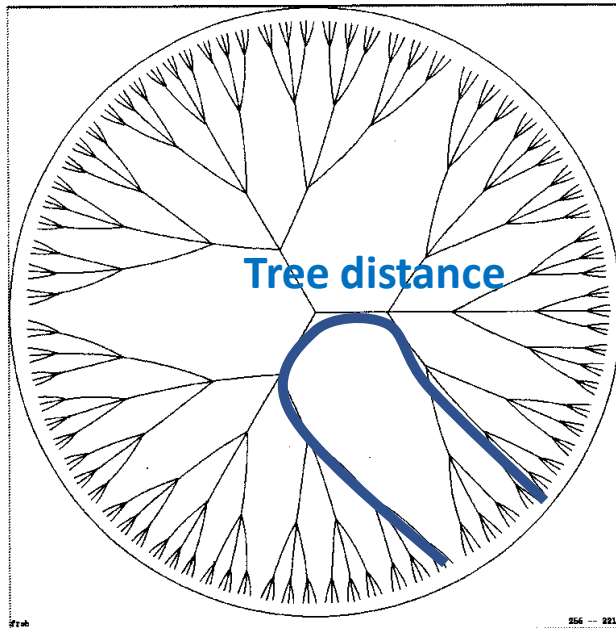
Poincaré: intuitive visualization



a hierarchical tree

Lower Distance Distortion

- In Poincaré model, geodesic bends inwards
- similar to trees: shortest path go through the LCA (lowest common ancestor)



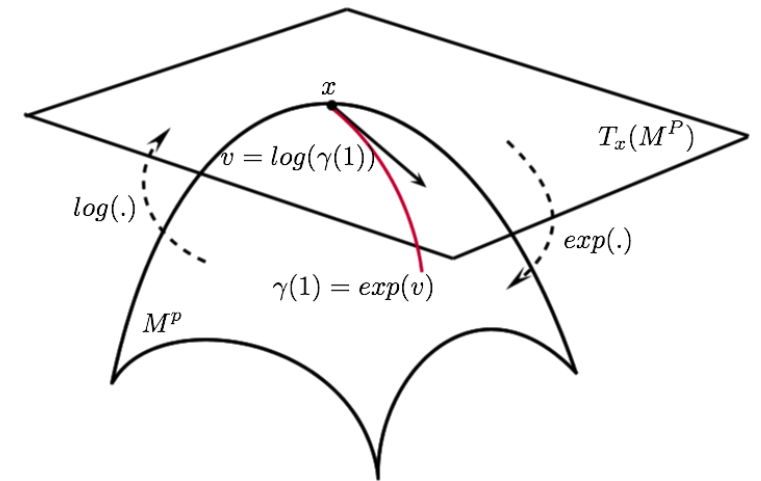
Content

- Non-Euclidean Space
- Hyperbolic Embeddings
- **Hyperbolic GNNs**

Challenges in Hyperbolic GNN

Challenges:

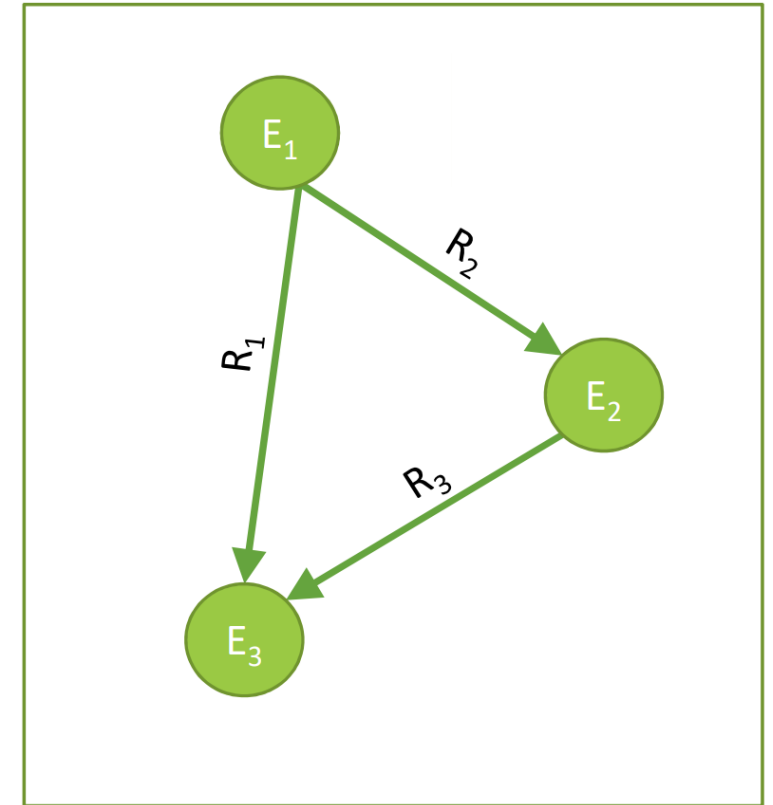
- Input node features are usually **Euclidean**
- Perform **hyperbolic aggregation** for message passing
- Choose hyperbolic spaces with the **right amount of curvature** at every layer of the GNN



Recap: Knowledge Graph

Knowledge in graph:

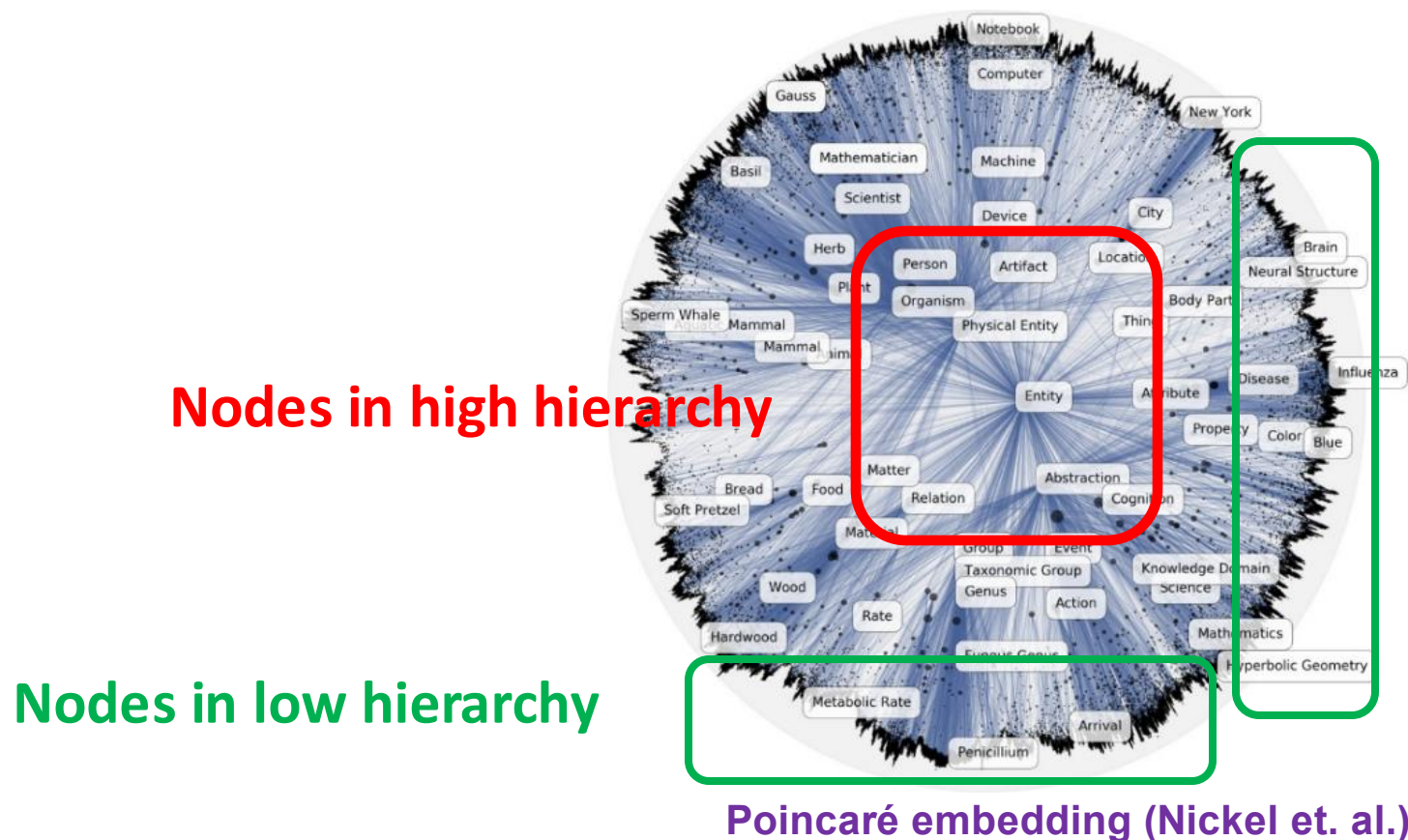
- A set of triplets <head entity, relationship, tail entity>
- Capture entities, types, and relationships
- Nodes are **entities**
- Nodes are labeled with their **types**
- Edges between two nodes capture **relationships** between entities
- **KG is an example of a heterogeneous graph**
 - **Recap:** Heterogeneous graph is a graph with multiple node types and edge types
- Commonly has tree-like structure



E: entity
R: relation

Tasks

- Graph representation learning on **hierarchical** graphs
 - Link Prediction
 - Node Classification



Problem Setting

- Given a Graph $G = (V, E)$, V is vertex set, E is edge set,
- $x_i^{0,E}$ indicates the **initial (first layer) feature** of node i in a Euclidean Space
- We use E to indicate the features in **Euclidean Space**, H to denote **hyperbolic space**, l to denote the **l -th layer** feature
- **Goal:** learn a mapping f which maps nodes to d -dimension embedding vectors

$$f: (V, E, (x_i^{0,E})_{i \in V}) \rightarrow Z \in \mathbb{R}^{|V| \times d}$$

Overview: Hyperbolic GNN (HGNCN)

- $\mathbf{h}_i^{l,H} = \text{Msg}(\mathbf{x}_i^{l-1,H})$
- $\mathbf{y}_i^{l,H} = \text{AGG}^{K_{l-1}}(\mathbf{h}_i^{l,H})$
- $\mathbf{x}_i^{l,H} = \text{Update}^{K_{l-1},K_l}(\mathbf{y}_i^{l,H})$

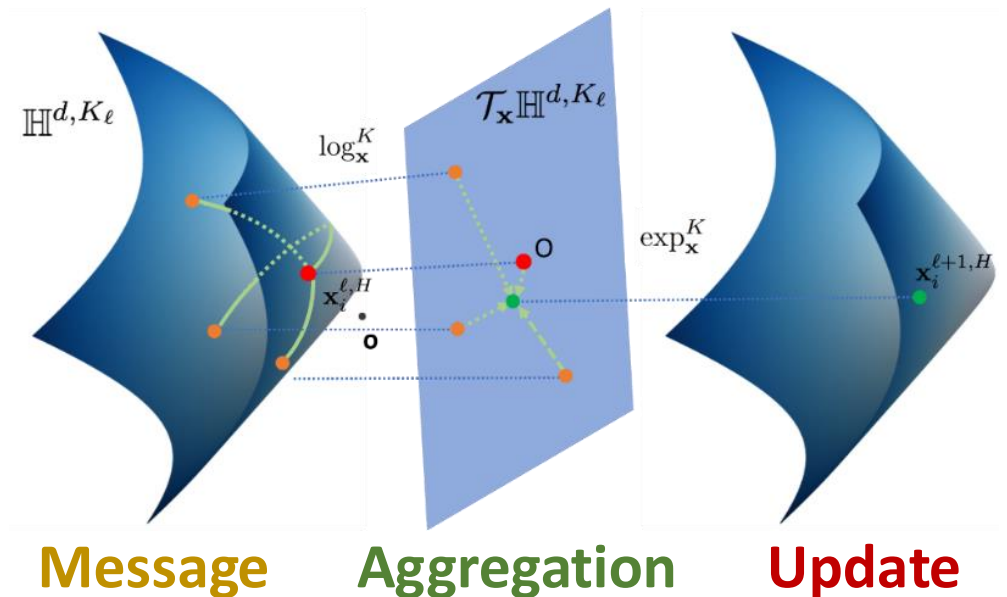
Message

Aggregation

Update

K_l : curvature at layer l

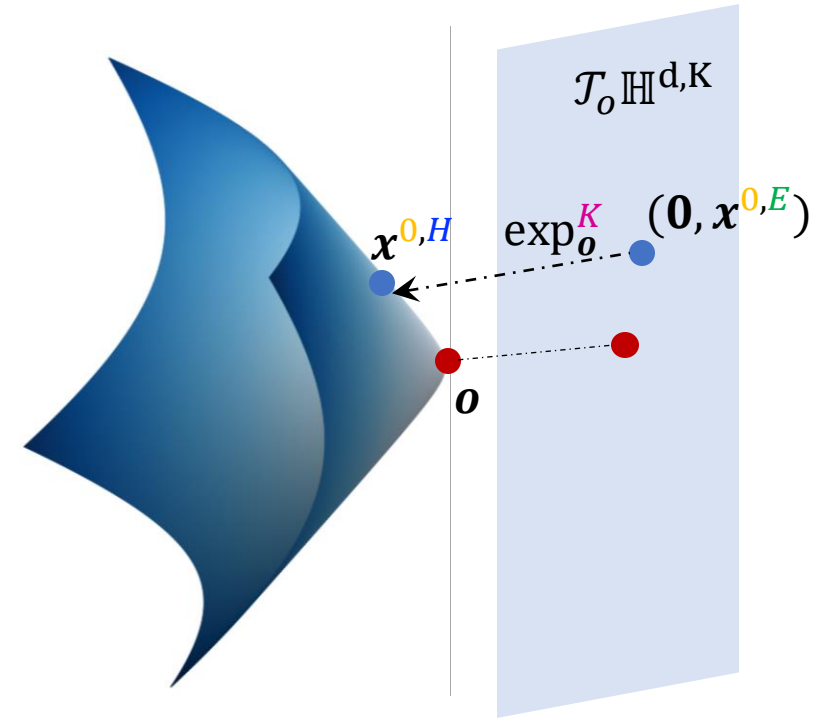
At every layer:



Hyperbolic GNN: Transformation

- $\mathbf{x}^{0,E} \in \mathbb{R}^d$ denotes input Euclidean features
- We use $\mathbf{o} = \{\sqrt{K}, \mathbf{0}, \dots, \mathbf{0}\} \in \mathbb{H}^{d,K}$ (the north pole in $\mathbb{H}^{d,K}$) as a reference point to perform exponential mapping
 - $\mathcal{T}_o \mathbb{H}^{d,K} = \{\mathbf{v} \in \mathbb{R}^{d+1} : \langle \mathbf{v}, \mathbf{o} \rangle_{\mathcal{L}} = 0\}$
 - We have $\langle \mathbf{o}, (0, \mathbf{x}^{0,E}) \rangle = 0$

$(0, \mathbf{x}^{0,E})$ can be interpreted as a point in $\mathcal{T}_o \mathbb{H}^{d,K}$!



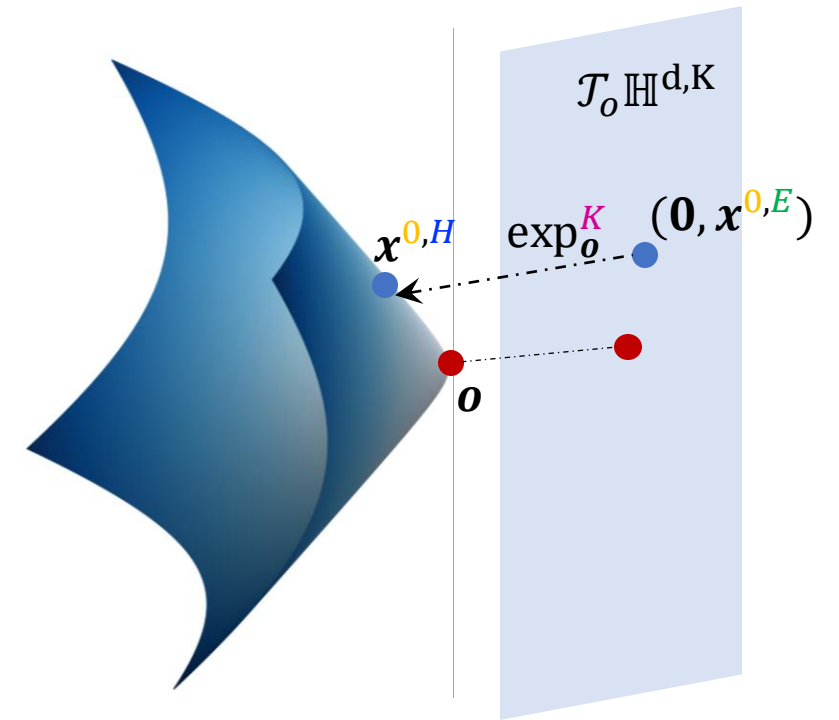
Hyperbolic GNN: Transformation

- **Input Transformation**

$$\mathbf{x}^{0,H} := \exp_o^K((0, \mathbf{x}^{0,E}))$$

$$= (\sqrt{K} \cosh\left(\frac{\|\mathbf{x}^{0,E}\|_2}{\sqrt{K}}\right), \sqrt{K} \sinh\left(\frac{\|\mathbf{x}^{0,E}\|_2}{\sqrt{K}}\right) \frac{\mathbf{x}^{0,E}}{\|\mathbf{x}^{0,E}\|_2})$$

- $(0, \mathbf{x}^{0,E})$ is a point in $\mathcal{T}_o \mathbb{H}^{d,K}$
- $\exp_o^K$ maps the point to $\mathbb{H}^{d,K}$



Hyperbolic GNN: Message

- **Message:**

$$h_i^{l,H} = (W^l \otimes^{K_{l-1}} x_i^{l-1,H}) \oplus^{K_{l-1}} b^l$$

- Hyperbolic linear: $W^l \otimes^K x^H := \exp_o^K (W^l \log_o^K (x^H))$
 - $\log_o^K$ maps hyperbolic points x^H to tangent space $\mathcal{T}_o \mathbb{H}^{d_1, K}$
 - do linear transformation in $\mathcal{T}_o \mathbb{H}^{d, K}$ with transformation matrix $W^l \in \mathbb{R}^{d_2 \times d_1}$
 - $\exp_o^K$ maps points back to the hyperboloid $\mathbb{H}^{d_2, K}$
- Mobius addition: $x^H \oplus^K b := \exp_{x^H}^K (P_{o \rightarrow x^H}^K(b))$

In tangent space $\mathcal{T}_{x^H} \mathbb{H}^{d, K}$

Recap: Parallel Transport

$$P_{x \rightarrow y}(v) = v - \frac{\langle \log_x^K(y), v \rangle_{\mathcal{L}}}{D_{\mathcal{L}}^K(x, y)^2} (\log_x^K y + \log_y^K x)$$

Hyperbolic GNN: Aggregation

- Given hyperbolic messages $\mathbf{h}_i^{l,H}, \mathbf{h}_j^{l,H}$ of node i and node j
- Map $\mathbf{h}_i^{l,H}, \mathbf{h}_j^{l,H}$ to the tangent space of the origin $\mathcal{T}_o \mathbb{H}^{d,K}$ and calculate **attention weight** w_{ij}^l (node j to node i)

$$w_{ij}^l = \text{Softmax}_{j \in \mathcal{N}(i)} (\text{MLP}(\log_o^K(\mathbf{h}_i^{l,H}) \parallel \log_o^K(\mathbf{h}_j^{l,H})))$$

- **Aggregation:**

$$\mathbf{y}_i^{l,H} = \text{AGG}^K(\mathbf{h}^{l,H})_i := \exp_{\mathbf{h}_i^{l,H}}^K \left(\sum_{j \in \mathcal{N}(i)} w_{ij}^l \log_{\mathbf{h}_i^{l,H}}^K(\mathbf{h}_j^{l,H}) \right)$$

- Note: curvature K is layer-wise and trainable!

Note: do aggregation in Tangent space $\mathcal{T}_{\mathbf{h}_i^{l,H}} \mathbb{H}^{d,K}$ when considering node i

Hyperbolic GNN: Update

- **Update:**

$$x_i^{l,H} = \text{Update}^{K_{l-1}, K_l}(\mathbf{y}_i^{l,H}) := \exp_o^{K_l}(\sigma(\log_o^{K_{l-1}}(\mathbf{y}_i^{l,H})))$$

- σ is a non-linear activation
- Apply activation in $\mathcal{T}_o \mathbb{H}^{d, K_{l-1}}$ and then map back to $\mathbb{H}^{d, K_l}$
- Tangent space of origin $\mathcal{T}_o \mathbb{H}^{d, K}$ is shared across hyperboloids $\mathbb{H}^{d, K}$ with any curvature K
- $\text{Update}^{K_{l-1}, K_l}(\cdot)$ enables HGNN to **smoothly vary curvature** at each layer from K_{l-1} to K_l

Hyperbolic GNN: Predict

- For **link prediction**, HGNCN uses Fermi-Dirac decoder:

$$p((i, j) \in E | x_i^{L,H}, x_j^{L,H}) = \left[e^{(d_L^{KL}(x_i^{L,H}, x_j^{L,H})^2 - r)/t} + 1 \right]^{-1}$$

- $p((i, j) \in E | x_i^{L,H}, x_j^{L,H}) \in (0, 1]$
- $d_L^{KL}(\cdot, \cdot)$ is the hyperbolic distance in $\mathbb{H}^{d, K_L}$
- r and t are hyper-parameters
- For **node classification**, use **logarithmic map** to map hyperbolic embeddings into Euclidean tangent space at O , and perform multi-class classification with standard softmax and cross entropy

δ -Hyperbolicity

Gromov's δ -Hyperbolicity

An undirected graph $G = (V, E)$ can be viewed as a metric space V with the graph distance d_G . Given $u, v, w, t \in V$ satisfying

$$d(u, v) + d(w, t) \geq d(u, t) + d(w, v) \geq d(u, w) + d(v, t),$$

we denote

$$\delta(u, v, w, t) = \frac{d(u, v) + d(w, t) - d(u, t) - d(w, v)}{2}$$

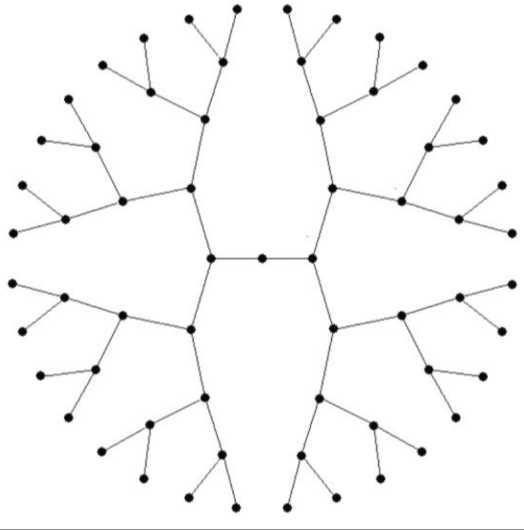
 **Four-points condition**

The δ -Hyperbolicity of the graph is defined as

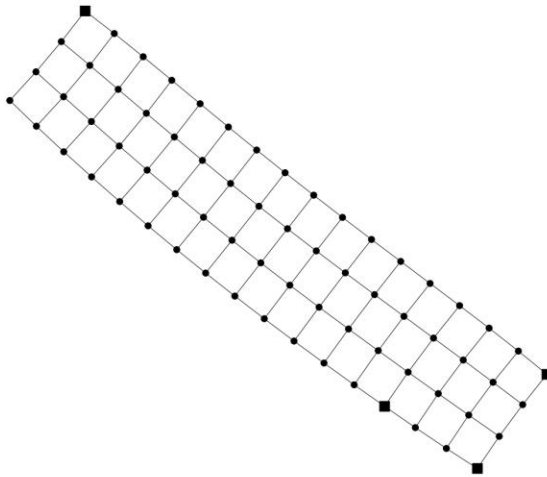
$$\delta(G, d_G) = \sup_{u, v, w, t \in V} \delta(u, v, w, t)$$

δ -Hyperbolicity

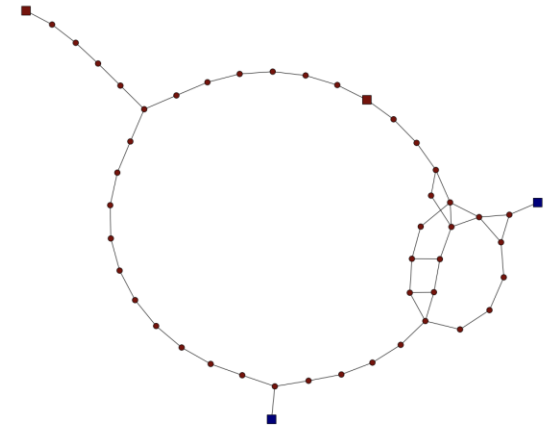
- The **lower** δ , the **more hyperbolic** is the graph
- $\delta = 0$ for trees.



$\delta = 0$



$\delta = 3.0$



$\delta = 4.5$

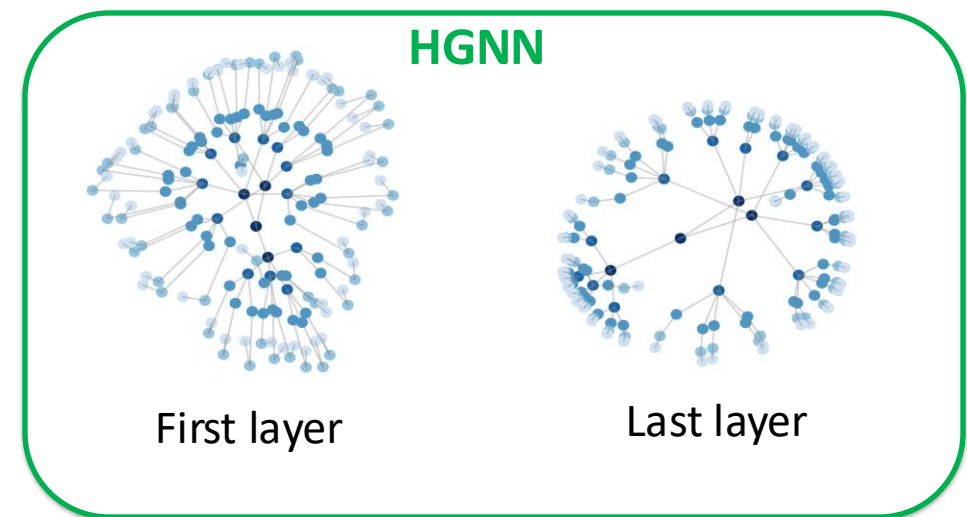
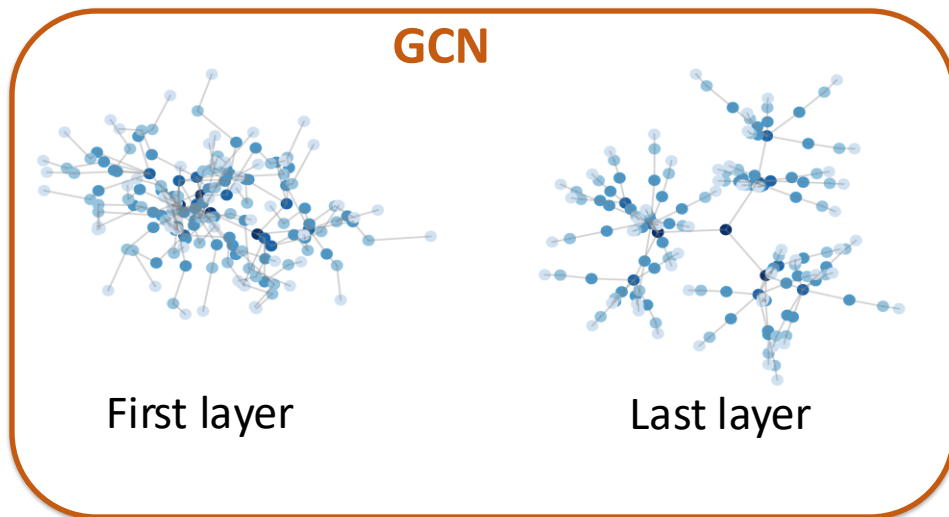
Experimental Results

Dataset		DISEASE		DISEASE-M		HUMAN PPI		AIRPORT		PUBMED		CORA	
Hyperbolicity δ		$\delta = 0$		$\delta = 0$		$\delta = 1$		$\delta = 1$		$\delta = 3.5$		$\delta = 11$	
Method		LP	NC	LP	NC	LP	NC	LP	NC	LP	NC	LP	NC
Shallow	EUC	59.8 $\pm$ 2.0	32.5 $\pm$ 1.1	-	-	-	-	92.0 $\pm$ 0.0	60.9 $\pm$ 3.4	83.3 $\pm$ 0.1	48.2 $\pm$ 0.7	82.5 $\pm$ 0.3	23.8 $\pm$ 0.7
	HYP [29]	63.5 $\pm$ 0.6	45.5 $\pm$ 3.3	-	-	-	-	94.5 $\pm$ 0.0	70.2 $\pm$ 0.1	87.5 $\pm$ 0.1	68.5 $\pm$ 0.3	87.6 $\pm$ 0.2	22.0 $\pm$ 1.5
	EUC-MIXED	49.6 $\pm$ 1.1	35.2 $\pm$ 3.4	-	-	-	-	91.5 $\pm$ 0.1	68.3 $\pm$ 2.3	86.0 $\pm$ 1.3	63.0 $\pm$ 0.3	84.4 $\pm$ 0.2	46.1 $\pm$ 0.4
	HYP-MIXED	55.1 $\pm$ 1.3	56.9 $\pm$ 1.5	-	-	-	-	93.3 $\pm$ 0.0	69.6 $\pm$ 0.1	83.8 $\pm$ 0.3	73.9 $\pm$ 0.2	85.6 $\pm$ 0.5	45.9 $\pm$ 0.3
NN	MLP	72.6 $\pm$ 0.6	28.8 $\pm$ 2.5	55.3 $\pm$ 0.5	55.9 $\pm$ 0.3	67.8 $\pm$ 0.2	55.3 $\pm$ 0.4	89.8 $\pm$ 0.5	68.6 $\pm$ 0.6	84.1 $\pm$ 0.9	72.4 $\pm$ 0.2	83.1 $\pm$ 0.5	51.5 $\pm$ 1.0
	HNN[10]	75.1 $\pm$ 0.3	41.0 $\pm$ 1.8	60.9 $\pm$ 0.4	56.2 $\pm$ 0.3	72.9 $\pm$ 0.3	59.3 $\pm$ 0.4	90.8 $\pm$ 0.2	80.5 $\pm$ 0.5	94.9 $\pm$ 0.1	69.8 $\pm$ 0.4	89.0 $\pm$ 0.1	54.6 $\pm$ 0.4
GNN	GCN[21]	64.7 $\pm$ 0.5	69.7 $\pm$ 0.4	66.0 $\pm$ 0.8	59.4 $\pm$ 3.4	77.0 $\pm$ 0.5	69.7 $\pm$ 0.3	89.3 $\pm$ 0.4	81.4 $\pm$ 0.6	91.1 $\pm$ 0.5	78.1 $\pm$ 0.2	90.4 $\pm$ 0.2	81.3 $\pm$ 0.3
	GAT [41]	69.8 $\pm$ 0.3	70.4 $\pm$ 0.4	69.5 $\pm$ 0.4	62.5 $\pm$ 0.7	76.8 $\pm$ 0.4	70.5 $\pm$ 0.4	90.5 $\pm$ 0.3	81.5 $\pm$ 0.3	91.2 $\pm$ 0.1	79.0 $\pm$ 0.3	93.7 $\pm$ 0.1	83.0 $\pm$ 0.7
	SAGE [15]	65.9 $\pm$ 0.3	69.1 $\pm$ 0.6	67.4 $\pm$ 0.5	61.3 $\pm$ 0.4	78.1 $\pm$ 0.6	69.1 $\pm$ 0.3	90.4 $\pm$ 0.5	82.1 $\pm$ 0.5	86.2 $\pm$ 1.0	77.4 $\pm$ 2.2	85.5 $\pm$ 0.6	77.9 $\pm$ 2.4
	SGC [44]	65.1 $\pm$ 0.2	69.5 $\pm$ 0.2	66.2 $\pm$ 0.2	60.5 $\pm$ 0.3	76.1 $\pm$ 0.2	71.3 $\pm$ 0.1	89.8 $\pm$ 0.3	80.6 $\pm$ 0.1	94.1 $\pm$ 0.0	78.9 $\pm$ 0.0	91.5 $\pm$ 0.1	81.0 $\pm$ 0.1
Ours	HGCN	90.8 $\pm$ 0.3	74.5 $\pm$ 0.9	78.1 $\pm$ 0.4	72.2 $\pm$ 0.5	84.5 $\pm$ 0.4	74.6 $\pm$ 0.3	96.4 $\pm$ 0.1	90.6 $\pm$ 0.2	96.3 $\pm$ 0.0	80.3 $\pm$ 0.3	92.9 $\pm$ 0.1	79.9 $\pm$ 0.2
	(%) ERR RED	-63.1%	-13.8%	-28.2%	-25.9%	-29.2%	-11.5%	-60.9%	-47.5%	-27.5%	-6.2%	+12.7%	+18.2%

- LP denotes link prediction
- NC denotes node classification

Embedding Visualization

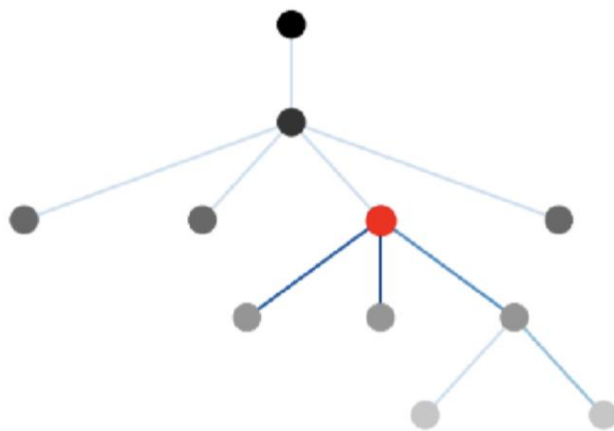
- Visualization on the Poincaré disk for link prediction on *DISEASE* ($\delta = 0$)
- Color indicates the **depth/hierarchy** of the node in a tree
 - Lighter color $\Rightarrow$ deeper in a tree $\Rightarrow$ lower hierarchy
- GCN hardly captures hierarchy, while HGNN preserves node hierarchies



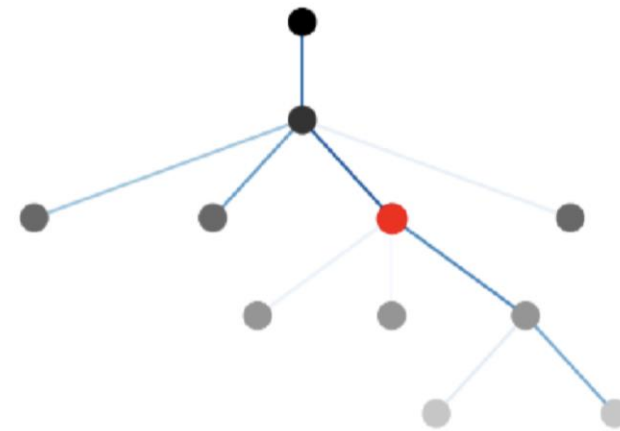
Lower distortion

Attention Visualization

- **Attention weights** in 2-hop neighbor of a center node on *DISEASE* ($\delta = 0$)
- **Darkness** of the color denotes their **hierarchy**. **Intensity** of the edges denotes the **attention weights**
- In HGNN, the center node pays more attention to its **(grand) parent**, who is with a **higher hierarchy**.



Euclidean GAT

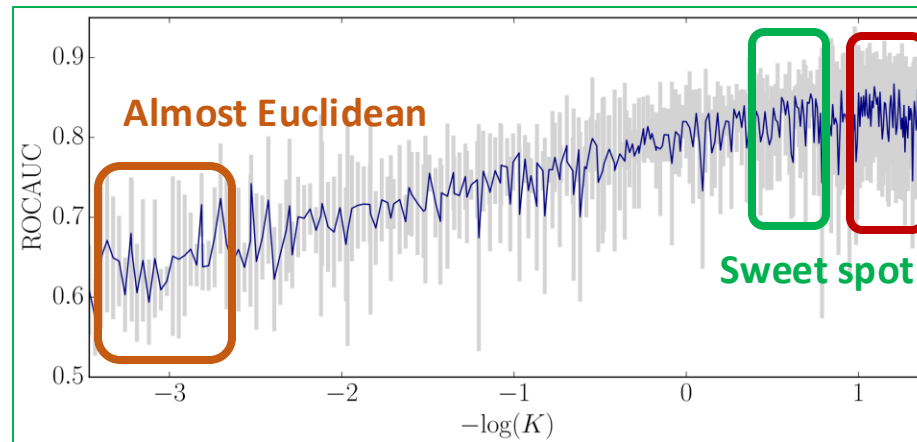


HGNN

● is the node where we compute attention

Performance V.S. Curvature

- Adjusting and training the curvature leads to improve the performance
- **Decreasing** the curvature **improves** link prediction performance on *DISEASE* ($\delta = 0$)



Too hyperbolic
(unstable)

Sweet spot

curvature $-\frac{1}{K}$ decreases

Curvature is important!

Summary of Hyperbolic Embedding

- **Hyperbolic embeddings** use hyperbolic geometry with constant negative curvature to preserve graph distances and complex relationships, particularly for **hierarchical and tree-like graphs**.
- **HGCN**: Graph convolutional network in **hyperbolic space**
 - maps Euclidean input features to hyperbolic embedding space, performs message aggregation in the tangent space and maps back to the hyperbolic space
- Experiments show decreasing the curvature of embedding space improves the performance over graphs with lower **δ -Hyperbolicity**.